

⅓ Country Reader

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Executive Summary

While overall progress occurs in the U.S., a paradox remains of deeply obscure, deepening internal divisions. To create a more granular view for effective policy, the ⅓ Country Reader introduces a novel framework that reimagines the U.S. as three distinct countries based on household income: Bottom Third (Low-Income), Middle Third (Medium-Income), and Top Third (High-Income). Utilizing data from sources like the U.S. Census and IPUMS, this framework evaluates how economic positioning shapes three critical societal domains: financial security and opportunity (Appendix A); health and human development (Appendix B); and equity, sustainability, and community resilience (Appendix C).

The reader aims to establish a detailed and integrated understanding of the American socioeconomic landscape, combining the three groups of statistical indicators with the three income groups and providing academic, industry, and government sources. This informs the current socioeconomic context of the U.S. using hundreds of indicators, explaining the internal divisions between the three income groups. This combination informs methods to foster more inclusive and equitable progress when distributing science, technology, and innovation. The report is structured into three sections:

- **Country Narratives:** Describes the Bottom Third facing severe economic stagnation, housing insecurity, and vulnerabilities in health, education, and infrastructure; the Middle Third experiencing moderate stability but increasing susceptibility to economic shifts and declining civic engagement; and the Top Third benefiting from robust economic security, superior infrastructure, and strong civic participation.
- **Visuals:** Detailed graphs map disparities across 60 income quantiles, clearly illustrating socioeconomic thresholds within and between groups. Notable visualizations center around Race/Ethnic Proportion, Educational Attainment, Housing Cost Burden, Labor Force Participation, Child School Enrollment, Household Crowding, Obesity Rate, Self-Rated Health Score, Non-Citizen Rate, and Voter Turnout.
- **Research Articles:** Collects academic, industry, and governmental research detailing structural inequalities in the key indicators of economic mobility, housing affordability, healthcare access, education quality, broadband access, civic engagement, and community resilience, highlighting areas critical for inclusive, equitable policy intervention.

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Introduction

The ⅓ Country Concept

Despite advancements in technology and economic growth, the U.S. faces lasting internal divisions. To understand these divisions, the U.S. typically assesses these problems using national averages, which can oversimplify widening socioeconomic disparities. To move beyond national averages, which obscure significant socioeconomic disparities, Dr. Michael Crow conceived the ⅓ Country Concept, dividing the U.S. into three distinct “countries” separated by income:

1. **Bottom Third Country (Low-Income):** Households that experience stagnation or declining socioeconomic conditions, typically due to the advancements in economic growth, technology, and innovation that are consolidated into higher income groups.
2. **Middle Third Country (Middle-Income):** Households that maintain moderate stability but are becoming increasingly vulnerable to changing economic conditions and developing socioeconomic pressures.
3. **Top Third Country (High-Income):** Households with historical socioeconomic prosperity that continues to grow, with substantial financial assets and robust access to technological and infrastructural developments.

This framework builds a more nuanced understanding of prosperity within multiple demographics within the U.S., charting disparities to better direct the national innovation system toward equitable progress for all Americans, as well as better informing how science funding and institutional structures can link basic research to solve complex problems.

Methodology

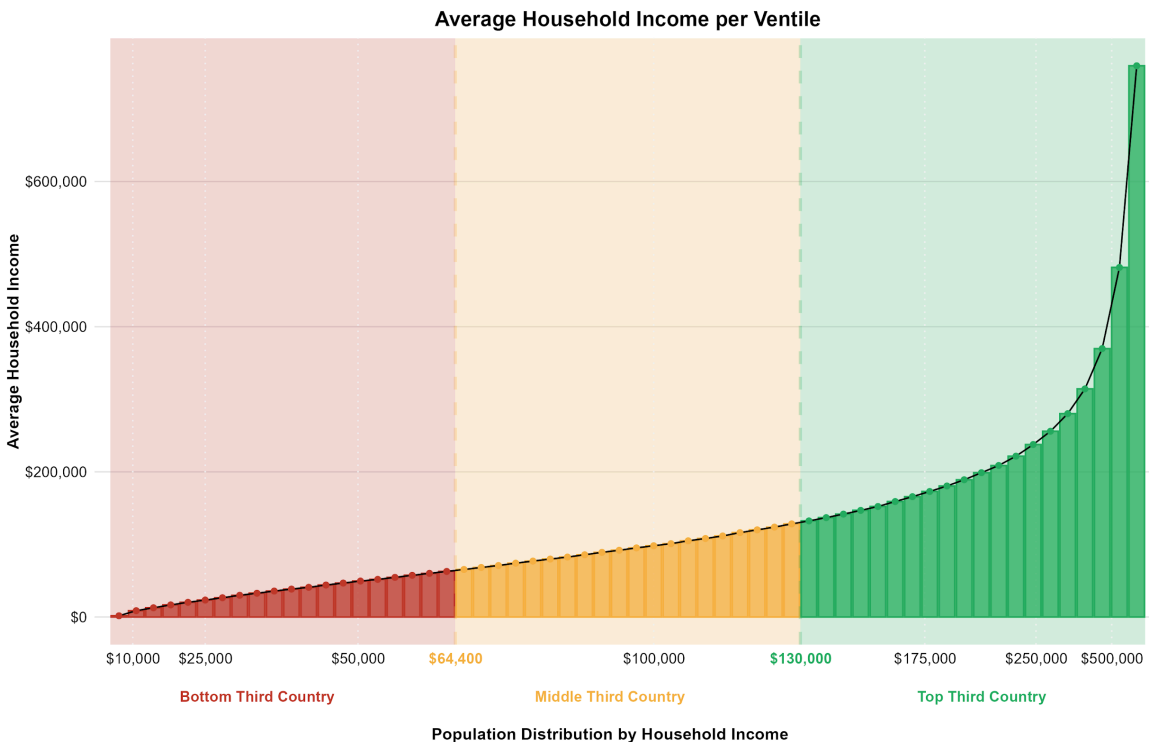
The ⅓ Country Concept aims to provide a nuanced understanding of diverse demographics and disparities within the U.S. To define the three thirds in terms of income levels, the report uses weighted household income data to categorize U.S. households into terciles (T):

1. **T1 (Low-Income)** includes households with incomes from approximately \$0 to \$64,400.
2. **T2 (Middle-Income)** includes households with incomes from around \$64,401 to \$130,000.
3. **T3 (High-Income)** includes households earning \$130,001 and above.

To analyze disparities with greater precision, each tercile is subdivided into 20 ventile groups, producing 60 quantile bins across the full income distribution. These segments were constructed using IPUMS ACS data with survey weights to capture income thresholds more precisely. For datasets that only report categorical income, such as certain CPS Supplements and the NHIS, continuous household income was imputed using CPS ASEC continuous household income distribution data, enabling consistent assignment to one of the 60 groups.

Figure 1 below shows the stratification of average household income per ventiles within the terciles. The figure involves mapping disparities across 60 income quantiles, illustrating socioeconomic thresholds within and between groups.

Figure 1: Average Household Income Per Ventile



Socioeconomic factors are then evaluated across three critical societal domains. These divisions allow comparison between the three levels of income based on a wide range of indicators of prosperity:

1. **Economic Security and Opportunity:** Assesses the financial stability, employment conditions, housing affordability, and upward mobility, as well as how economic resources, access to stable employment, and opportunities for advancement vary based on household income.
2. **Health and Human Development:** Focuses on disparities in healthcare access, quality of medical care, educational attainment, child and adult well-being, and overall life expectancy, as well as how economic standing directly impacts health outcomes, educational success, and lifelong human capital development.
3. **Equity, Sustainability, and Community Resilience:** Examines civic engagement, infrastructure quality, environmental resilience, access to essential resources such as broadband, food and water, and overall community cohesion, as well as how socioeconomic status shapes household's abilities to respond to challenges such as climate change, crime, and weak infrastructure.

Conclusion

By comparing indicators within these three income-based “countries”, the report explores the disparities in prosperity, well-being, and resilience across America’s socioeconomic spectrum. Using this visual methodology, which divides households into three income-based “countries” and further disaggregates them into ventiles, the report presents disparities in prosperity, well-being, and resilience across the socioeconomic spectrum. This structure enables consistent comparison across indicators while drawing from a broad array of datasets, including IPUMS ACS, CPS supplements, and NHIS. Grounded in literature that identifies income as a key determinant of access and outcome, this framework supports a deeper exploration of how economic positioning systematically shapes opportunity and outcomes in the U.S.

Country Narrative

This section uses a framework based on income terciles (Bottom, Middle, and Top Thirds) to show how prosperity differs across the income distribution. It draws from indicators grouped into three divisions—Economic Security and Opportunity, Health and Human Development, and Equity, Sustainability, and Community Resilience—each grounded in literature that identifies income as a key determinant of access and outcomes. This approach allows for analysis of distinct conditions: economic stagnation in the Bottom Third, vulnerabilities in the Middle Third, and advantage in the Top Third. These narratives provide context for understanding how each income group experiences opportunity and vulnerability, highlighting the systemic nature of inequality and its implications for national resilience and inclusive growth.

Bottom Third

In the Bottom Third, economic security is precariously balanced on the edge of crisis, significantly shaped by housing affordability and economic stagnation. Housing affordability emerges as a critical indicator, with nearly half of all renters facing severe housing cost burdens, spending over 30% of their income on housing alone. This financial strain hampers not only quality of life but also broader economic growth by restricting labor mobility, decreasing consumer spending, and deterring business investment. Households trapped by housing costs are unable to move to opportunity-rich regions, reducing their economic potential and reinforcing persistent regional disparities. Compounding these issues, economic mobility has drastically declined. Once a cornerstone of the American Dream, absolute income mobility—the likelihood that children earn more than their parents—has plummeted from over 90% for children born in the 1940s to around 50% for those born in the 1980s. This decline particularly affects the Bottom Third, where economic stagnation and structural inequality severely limit upward movement across generations. Additionally, unemployment disproportionately harms the Bottom Third, significantly reducing real incomes and intensifying economic misery. Unlike middle- or upper-income groups, where inflation and market fluctuations have less severe immediate impacts, the Bottom Third experiences amplified economic vulnerability due to employment instability and minimal financial buffers.

Health disparities within the Bottom Third are profound, deeply intertwined with economic conditions. Persistent inequities in healthcare access exacerbate vulnerabilities, particularly evident during public health crises like the COVID-19 pandemic. Rural communities and marginalized groups within the Bottom Third face chronic shortages of healthcare providers, limited telehealth access, and overall poorer healthcare outcomes. Structural inequities lead to higher chronic disease prevalence, diminished life expectancy, and compounded intergenerational health disadvantages. Education further compounds these disadvantages. Despite increased educational investment, large segments of students within the Bottom Third remain unable to meet basic academic standards. Achievement gaps persist, notably for Black and Hispanic students, limiting their pathways to higher education and better-paying jobs. Unequal access to high-quality education perpetuates cycles of poverty, restricting economic mobility and dampening long-term prosperity prospects.

Infrastructure underinvestment disproportionately impacts the Bottom Third, reinforcing economic disparities. Inadequate transportation and broadband infrastructure significantly constrain economic and social opportunities. Access to reliable transportation is particularly critical, especially in rural and small metropolitan areas where private vehicle ownership remains a primary determinant of employment access. Environmental vulnerabilities further exacerbate inequities. Climate change disproportionately threatens the Bottom Third, amplifying exposure to extreme weather, flooding, and pollution. Adaptation strategies often neglect lower-income regions, prioritizing economically affluent areas, thereby perpetuating existing inequalities and undermining long-term community resilience. Community resilience within the Bottom Third is also weakened by diminished civic engagement driven by inequality. Rising socioeconomic disparities correlate with reduced civic participation, lower social trust, and weakened community efficacy, severely limiting these communities' abilities to advocate effectively for policy and resource allocation.

The complex interplay of economic fragility, limited health and educational opportunities, and infrastructural and environmental vulnerabilities in the Bottom Third fundamentally challenges broader national prosperity. When substantial segments of the population face such systemic barriers, it restricts aggregate economic growth and stability, weakening national resilience. Prosperity, therefore, cannot be merely defined by aggregate GDP growth or national averages in health and education outcomes; it must explicitly incorporate equity, opportunity, and inclusiveness. Addressing these disparities requires integrated, multi-dimensional policy approaches aimed at dismantling structural barriers, improving access to affordable housing and quality education, investing equitably in infrastructure and healthcare, and prioritizing climate resilience strategies tailored to vulnerable communities. Prosperity in the Bottom Third—and consequently for the nation as a whole—depends upon targeted interventions that foster sustainable, inclusive economic growth and robust community resilience.

Middle Third

In the Middle Third, economic security is moderately stable but increasingly vulnerable to economic fluctuations such as inflation and macroeconomic shifts. This group experiences housing affordability issues less severely than the Bottom Third but still faces significant pressures, particularly in rapidly growing metropolitan areas. Homeownership, while achievable for many, remains precarious due to high property costs and interest rates, impacting long-term financial security. Income mobility within the Middle Third has noticeably declined, reflecting a broader erosion of upward mobility. Families find that traditional pathways to economic advancement, such as education and stable employment, do not guarantee upward movement as reliably as in previous decades. This diminished mobility affects confidence in sustained prosperity and highlights the necessity of economic strategies that secure long-term stability and equitable growth. Employment conditions for the Middle Third are generally stable, but job quality varies significantly. Inflation emerges as a significant stressor, disproportionately impacting middle-income families due to their reliance on stable incomes and limited capacity

for financial cushioning through asset diversification. Consequently, economic policy and labor market conditions profoundly influence the resilience and prosperity of the Middle Third.

The Middle Third typically enjoys relatively consistent healthcare access, though disparities remain, particularly regarding affordability and comprehensive coverage. Healthcare outcomes are broadly positive but exhibit vulnerabilities, particularly during economic downturns or health crises that strain personal and public healthcare resources. Preventive care and chronic disease management are accessible yet remain sensitive to fluctuations in healthcare costs and insurance coverage stability. Educational attainment within the Middle Third is comparatively high, providing significant individual economic mobility and stability. However, challenges persist, including educational debt and uneven returns on educational investments, especially affecting younger generations. Persistent racial and gender wage disparities, despite similar education levels, underscore the ongoing need for equitable economic and educational reforms to ensure broad-based prosperity.

Infrastructure quality within the Middle Third varies widely, typically better than that in the Bottom Third but often falling short of the resources and investments accessible to higher-income communities. Transportation and digital connectivity are generally adequate but frequently lag rapidly changing technological and infrastructural standards. Ensuring robust, forward-looking investments in infrastructure remains essential for enhancing economic competitiveness and community resilience. Climate resilience is increasingly critical for the Middle Third, given their economic vulnerability to environmental disruptions. While possessing greater adaptive capacity than lower-income groups, middle-income communities still face significant risks from extreme weather events and other climate-related disruptions, which can quickly erode economic security and community stability. Civic engagement in the Middle Third remains relatively robust, facilitating effective community advocacy and social cohesion. However, socioeconomic pressures and increasing inequality could potentially diminish civic participation if not adequately addressed, highlighting the importance of policies promoting inclusive growth and community empowerment.

For the Middle Third, prosperity involves balancing economic stability with sustained growth opportunities. Ensuring prosperity requires addressing structural vulnerabilities such as housing affordability, employment quality, healthcare affordability, and educational debt. Investments in infrastructure and equitable access to resources are crucial for maintaining economic stability and enhancing resilience against economic and environmental shocks. Strategically addressing these factors through comprehensive policy frameworks—prioritizing economic stability, educational equity, infrastructure resilience, and climate adaptation—will reinforce prosperity. Prosperity for the Middle Third thus necessitates continuous commitment to equitable policies and robust social infrastructure that collectively enable sustained, inclusive economic growth and social stability.

Top Third

In the Top Third, economic security and opportunity are robust, characterized by high homeownership rates, substantial financial assets, and strong labor market positions. Housing affordability is a minimal concern due to higher disposable incomes and significant asset holdings, allowing individuals greater flexibility in choosing residential locations aligned with optimal economic opportunities and quality of life. Income mobility within the Top Third remains high, supported by ample resources and access to premier education and professional networks. This group's economic advancement is facilitated by substantial generational wealth transfer and sustained asset appreciation, making intergenerational mobility less reliant on labor income alone and more secure against economic volatility. Employment conditions are markedly stable, characterized by high-quality job opportunities, significant bargaining power, and comparatively minimal vulnerability to unemployment. Inflation has limited immediate impact on the Top Third, as financial strategies and asset diversification provide effective hedging against economic fluctuations.

The Top Third enjoys comprehensive and premium healthcare access, significantly exceeding national averages in both preventative and specialty care. Health outcomes are consistently superior due to extensive healthcare resources, advanced medical facilities, and robust insurance coverage, resulting in higher life expectancy and better overall health profiles. Educational attainment within the Top Third is exceptionally high, closely linked to sustained economic prosperity. Members of this group have privileged access to top-tier educational institutions, extensive support networks, and significantly lower educational debt burdens, reinforcing their economic advantage. However, ongoing disparities in educational outcomes for women and racial minorities within this tier highlight continuing equity challenges.

Infrastructure within the Top Third is exceptionally well-developed, encompassing superior transportation networks, extensive digital connectivity, and robust public amenities. This infrastructure not only enhances daily quality of life but also significantly bolsters economic productivity and innovation capacity. Climate resilience for the Top Third is high, supported by substantial adaptive capacity, superior infrastructure, and ample financial resources to mitigate and respond to environmental threats. However, this adaptive advantage raises critical equity issues, as climate adaptation efforts often prioritize affluent communities, reinforcing systemic disparities in environmental vulnerability and resilience. Civic engagement is strong among the Top Third, driven by higher levels of education, greater economic security, and extensive access to political and social networks. While this active participation generally promotes community stability and influence over policy outcomes, it also raises concerns about disproportionate political representation and potential marginalization of lower-income groups.

Prosperity within the Top Third is robust and multidimensional, reflecting substantial economic stability, superior human development outcomes, and exceptional community resilience. Yet, the concentrated prosperity of this segment underscores the broader societal challenges posed by widening inequality. Sustained prosperity requires addressing disparities through equitable policy frameworks aimed at extending similar opportunities and stability to middle- and lower-income groups. Ensuring long-term national prosperity thus involves proactive engagement by the Top Third in policies promoting equitable economic growth,

enhanced social mobility, and inclusive community development. This engagement is essential not only for societal cohesion but also for maintaining a dynamic, resilient, and prosperous nation

Visuals

Figure 2: Age Structure

- Average age exhibits a U-shaped pattern. It is highest in the lower-middle part of the Bottom Third (around the \$25k mark, quantile positions 5-10), decreases through the Middle Third, reaches its lowest point in the lower part of the Top Third (around \$130k-\$175k, positions 41-50), and then rises again at the very top incomes.
- This suggests that different life-cycle stages are concentrated at different income levels.

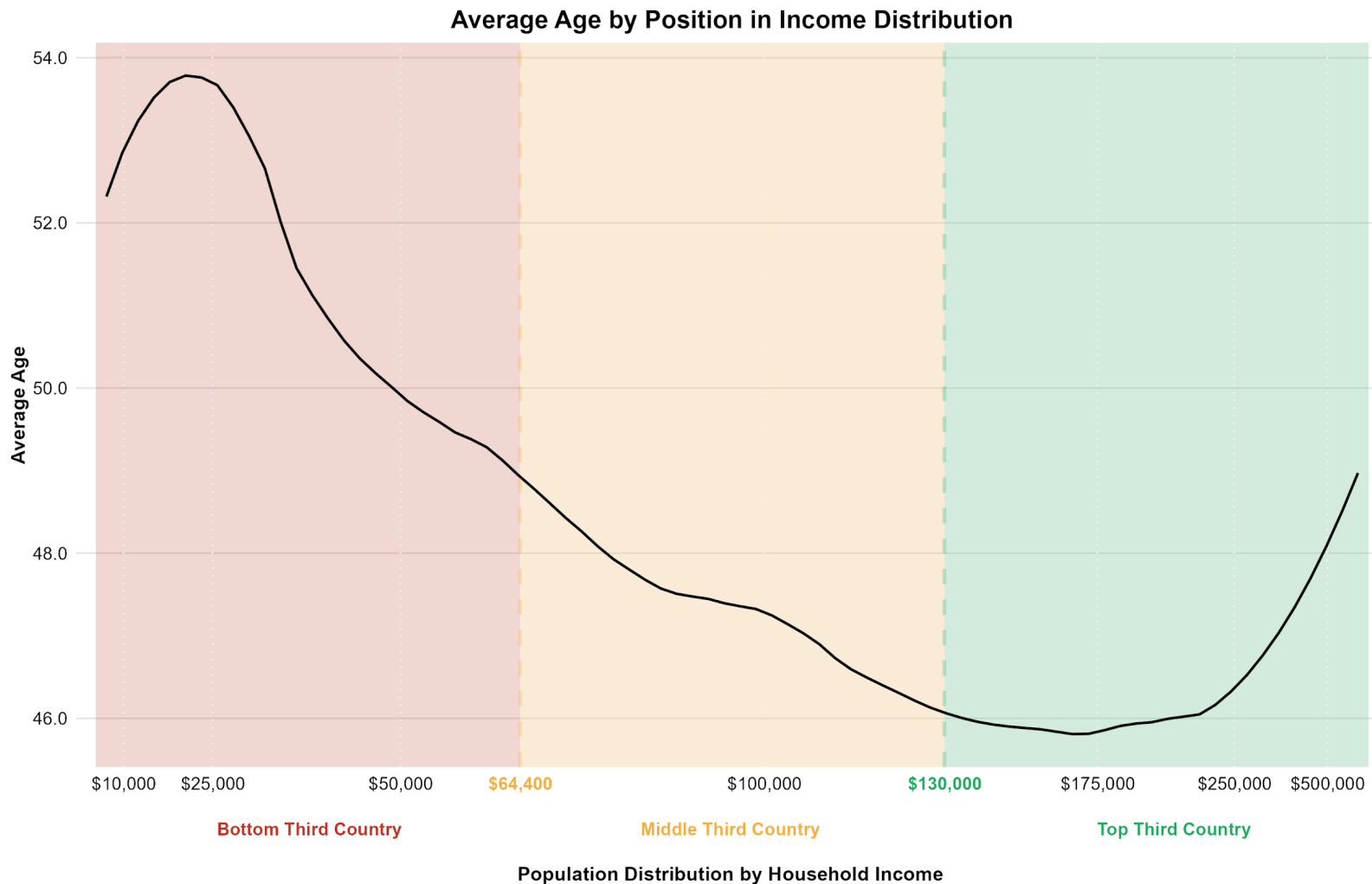


Figure 3: Racial/Ethnic Composition

- Significant shifts occur across the distribution. The proportion identifying as White, Non-Hispanic steadily increases with income, becoming the clear majority (over 60%) in the Top Third. Conversely, the proportion identifying as Hispanic is highest in the Bottom Third (around 20–23%) and declines steadily through the Middle and Top Thirds.
- The proportion identifying as Black, Non-Hispanic peaks in the lower-middle part of the Bottom Third (around 15%) and gradually decreases at higher income levels.

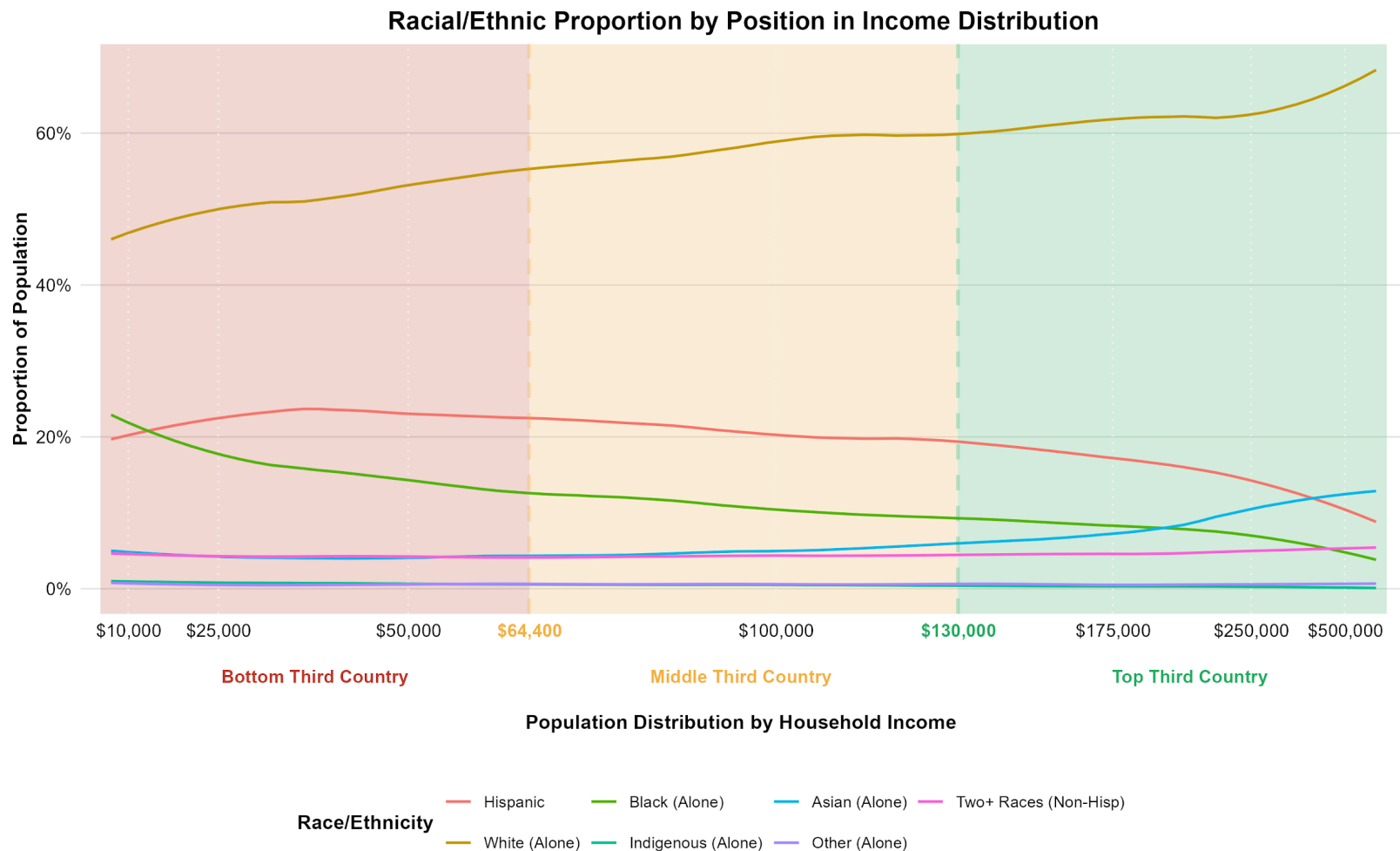


Figure 4: Racial Diversity

- The Racial Diversity Index (calculated as the Herfindahl-Hirschman Index based on the proportions in A3) is highest at the lowest income levels (around 0.68–0.70), indicating greater racial/ethnic heterogeneity.
- Diversity consistently decreases as income rises, reaching its lowest point (around 0.5) at the very top, reflecting the increasing predominance of the White, non-Hispanic group at higher income levels.

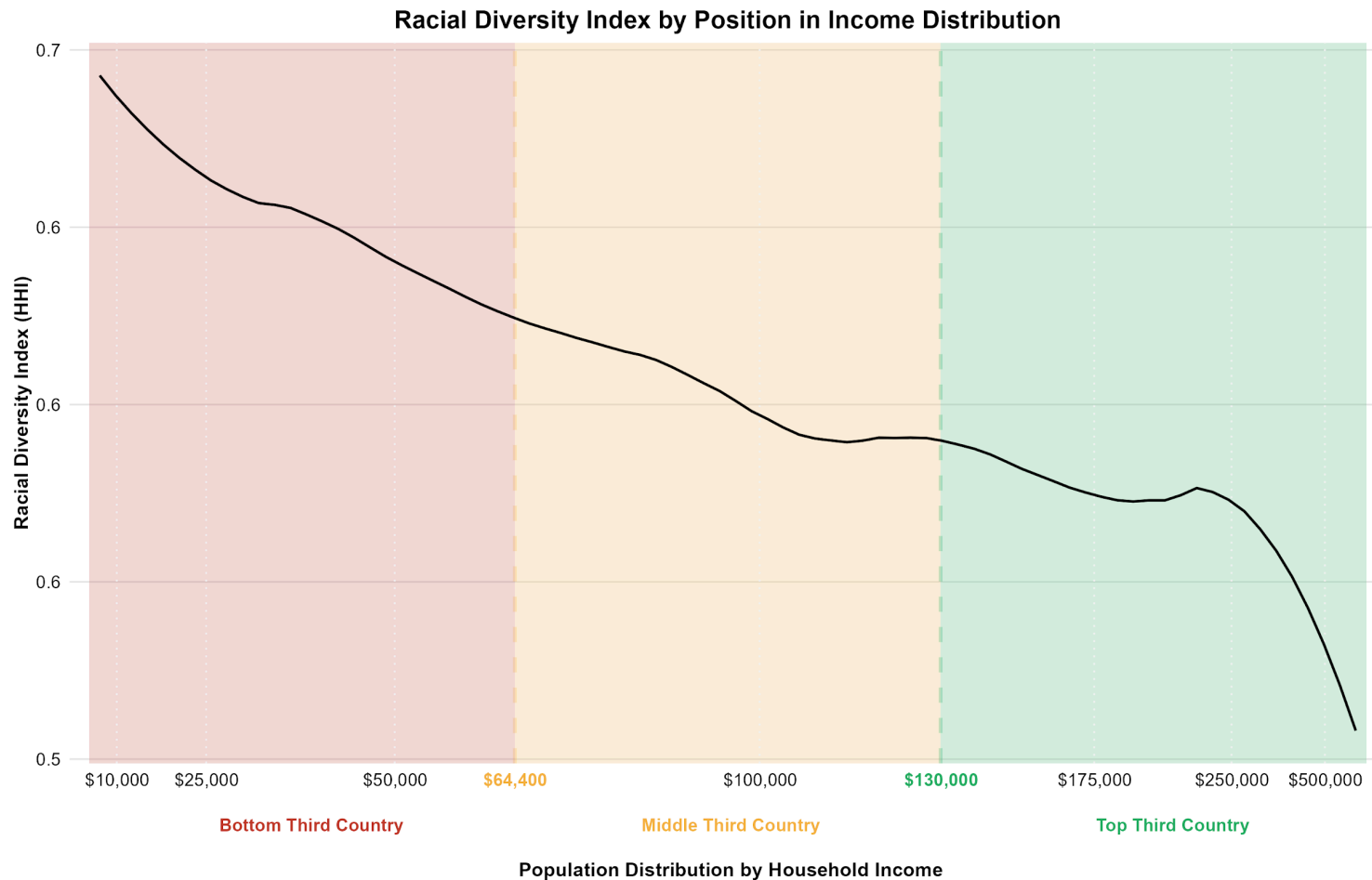


Figure 5: Labor Force and Employment

- Labor Force Participation (LFP) and Employment Rates (among ages 25-65, left axis) show a strong positive correlation with income position. Both rates rise steeply through the Bottom Third, continue to increase more gradually through the Middle Third, and plateau at high levels (around 85-90%) in the Top Third.
- Conversely, the U-3 Unemployment Rate (right axis) exhibits a sharp inverse relationship, plummeting from over 20% at the lowest incomes to below 5% through most of the Middle and Top Thirds.



Figure 6: Poverty

- Within the Bottom Third country (income < \$64.4k), both overall and child poverty rates decrease dramatically as income position increases.
- At the very lowest ventiles (below \$10k-\$15k), poverty rates approach 100%. The decline is steep through the \$15k-\$40k range, eventually reaching near-zero levels by the Bottom Third and Middle Third border. Child poverty rates are consistently higher than overall poverty rates throughout this tercile.



Figure 7: Food Insecurity

- The rate of food insecurity (based on CPS FSS) shows a strong negative correlation with income. It is highest (over 35%) at the lowest income ventiles and decreases sharply through the Bottom Third.
- The decline continues, albeit more gradually, through the Middle Third, reaching low single digits in the Top Third.

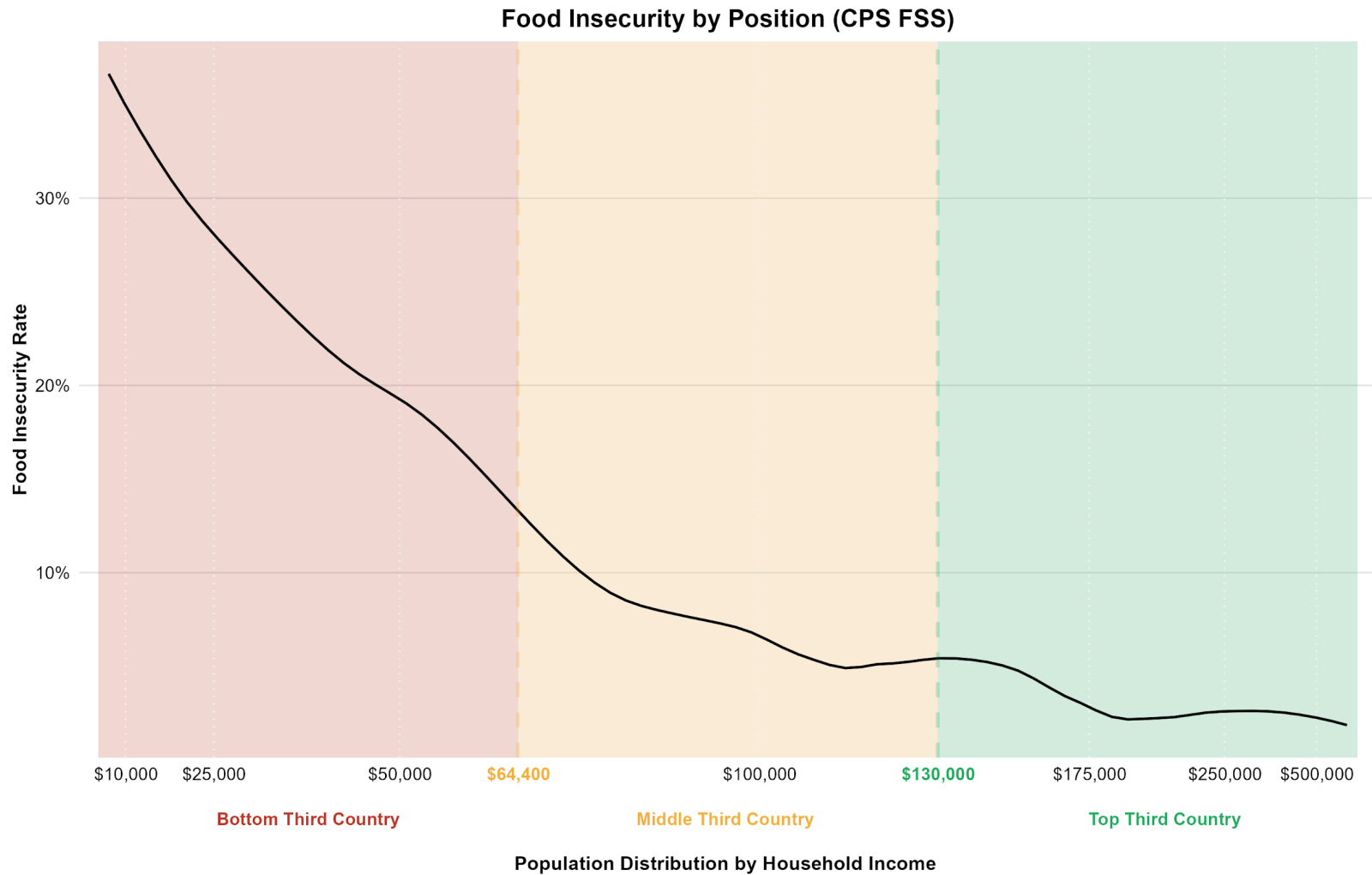


Figure 8: Socioeconomic Status

- Average HWSEI (Hauser-Warren Socioeconomic Index, based on occupation) displays a distinct pattern.
- It initially decreases slightly at the very bottom incomes (suggesting low-income workers in slightly higher-status occupations are not at the absolute bottom of the income distribution), then rises steadily and significantly across the rest of the Bottom, Middle, and Top Thirds, indicating a strong link between occupational status and household income.

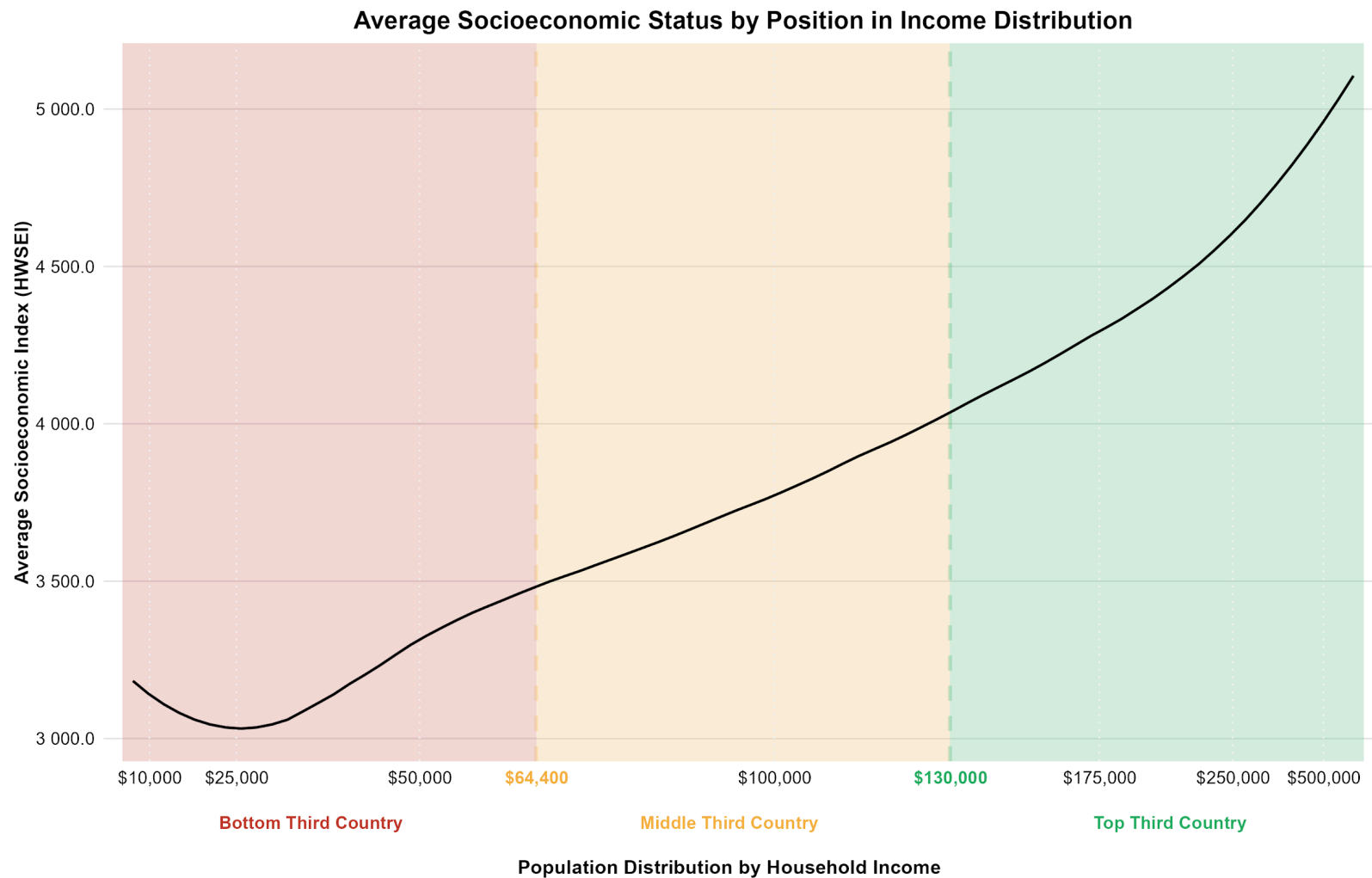


Figure 9: Educational Attainment

- The proportion of adults (25 years old+) with a High School degree or less is highest (over 80%) at the lowest income levels and declines steadily across the distribution, falling below 25% in the upper range of the Top Third. Conversely, the proportion with a Bachelor's degree increases steadily from around 10% in the Bottom Third to nearly 40% in the Top Third.
- The proportion with a Graduate degree shows the steepest increase, rising from under 5% in the Bottom Third to over 40% at the highest income levels, surpassing the bachelor's-only group.

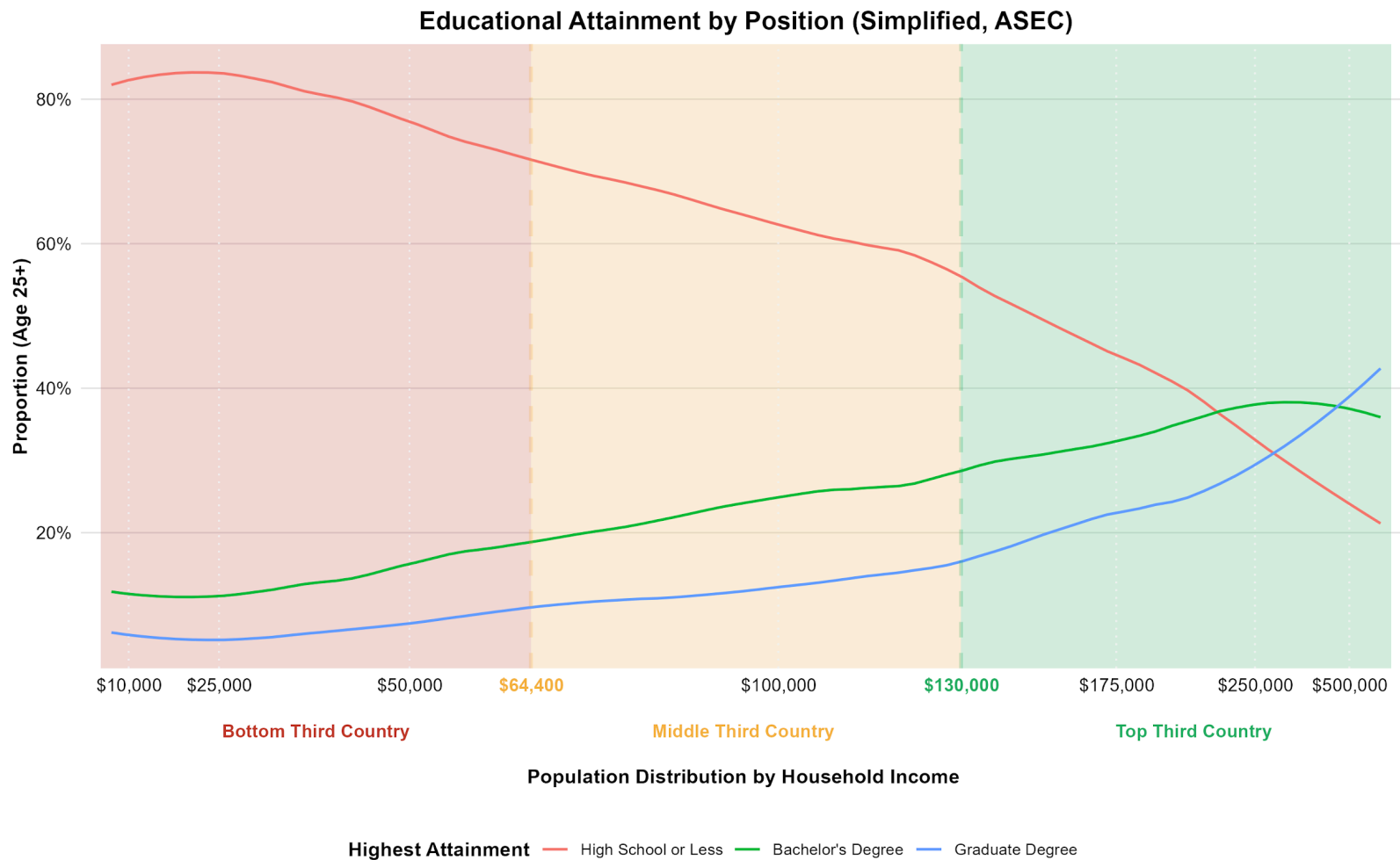


Figure 10: Child Enrollment

- School enrollment rates for children (<18) generally increase with income position.
- There's a sharp rise in the lower Bottom Third, a dip and recovery around the middle of the Bottom Third, followed by a steadier increase through the Middle Third, reaching near-universal enrollment (>95%) in the Top Third.

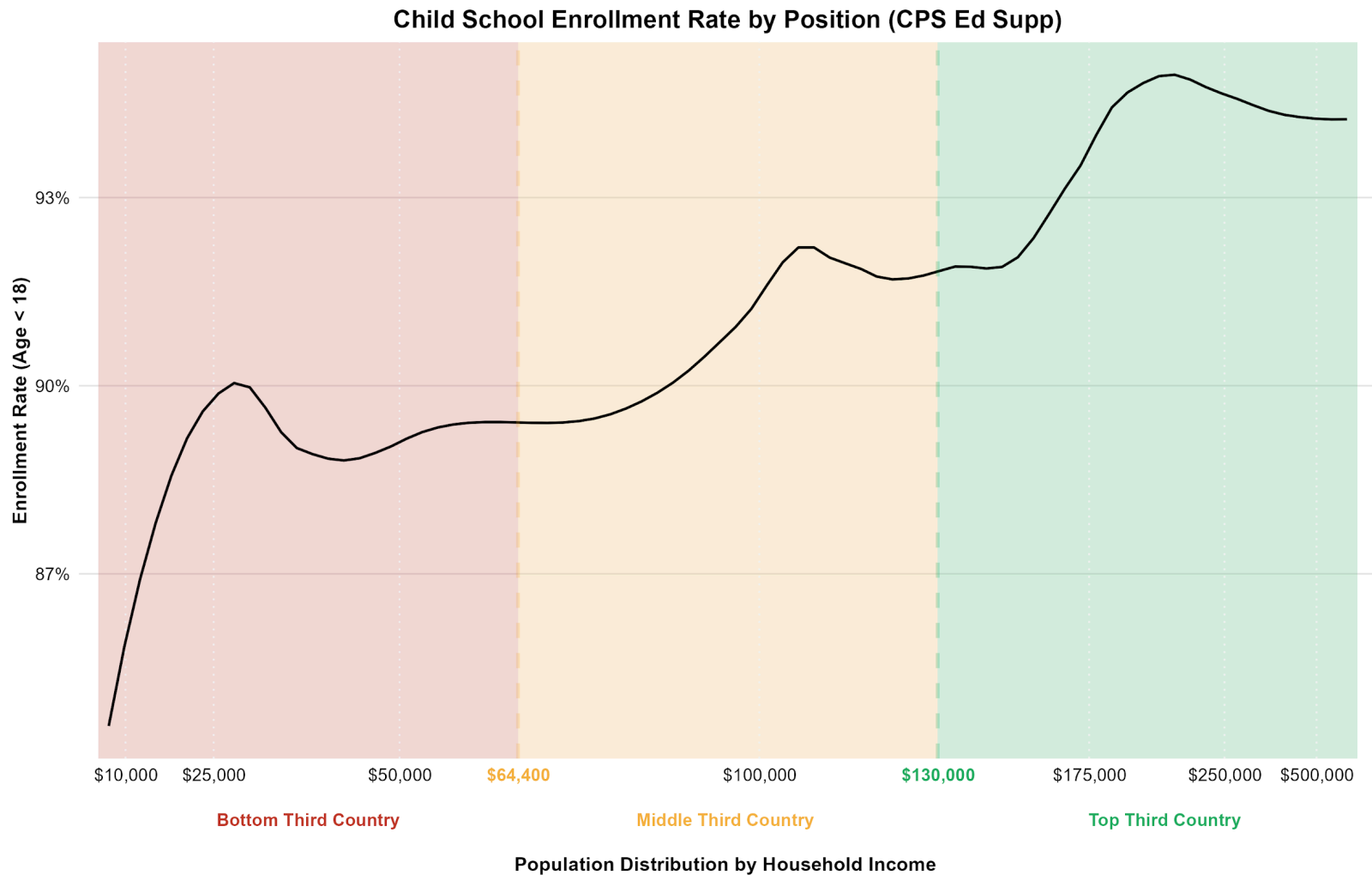


Figure 11: Adult Enrollment

- Formal education enrollment among adults (25 years old+) shows a more complex pattern. It is highest (>5%) at the very lowest income levels, decreases significantly in the rest of the Bottom Third, and then rises again through the Middle Third, peaking around the Middle Third and Top Third border before declining gradually through the Top Third.
- This suggests different motivations or types of enrollment may be prevalent at different income levels (e.g., retraining at lower levels, advanced degrees in the middle/upper-middle).

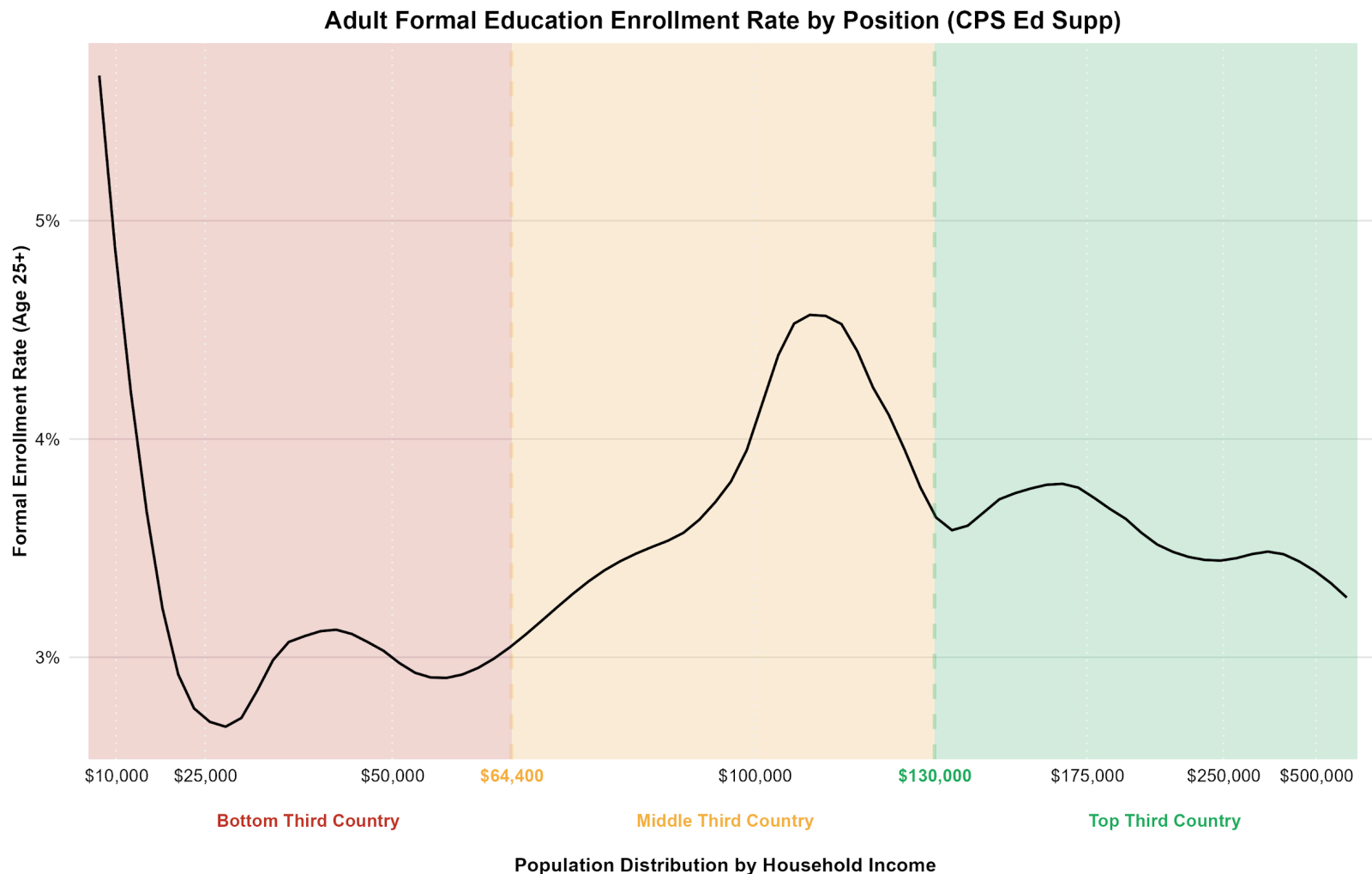


Figure 12: Housing Tenure

- Homeownership rates increase dramatically with income position, while rentership rates decrease correspondingly.
- Ownership starts below 40% at the lowest incomes, surpasses rentership around the \$40k mark within, and exceeds 80-90% through most of the Top Third.

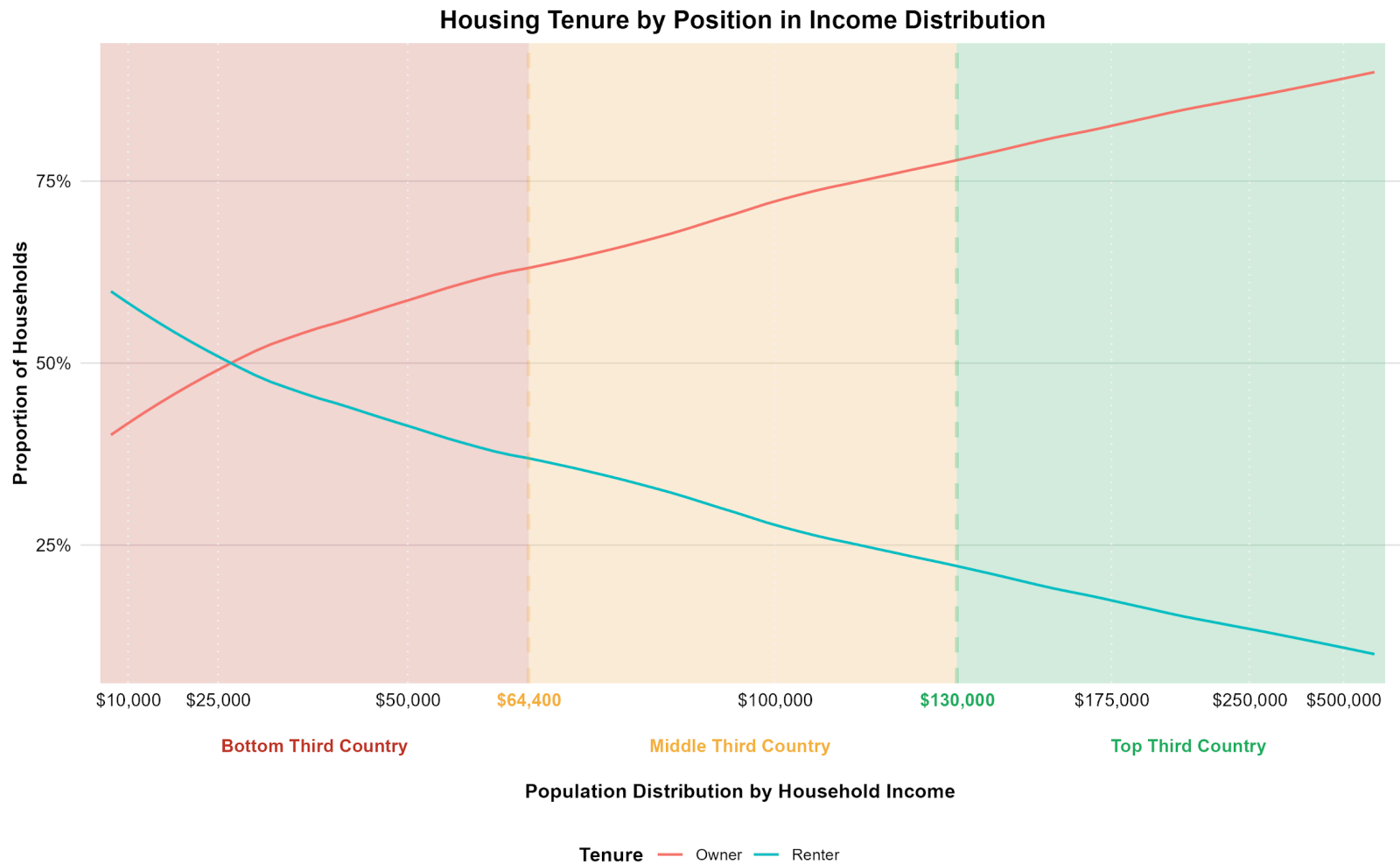


Figure 13: Housing Cost Burden

- Cost burden (spending >30% of income on housing) shows different patterns for owners and renters. Renter cost burden is extremely high (over 40%) in the lower Bottom Third and decreases sharply with income.
- The owner cost burden starts lower (around 30%) but decreases more gradually. Renters are more likely to be burdened than owners below roughly \$55k (mid- Bottom Third), while owners are slightly more likely to be burdened above that point, although rates for both become very low in the Top Third.

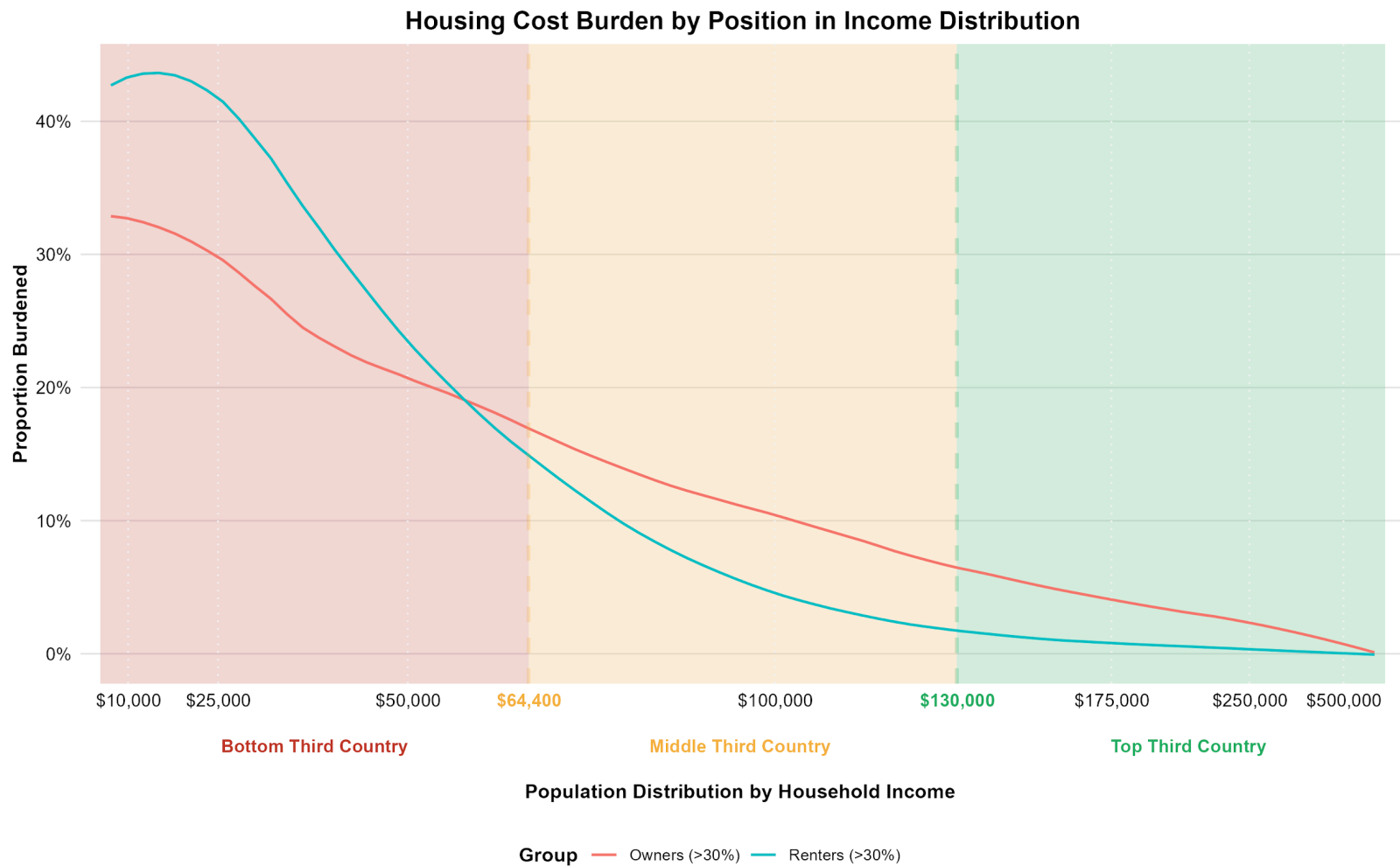


Figure 14: Household Crowding

- Average persons per bedroom decreases sharply as income rises, particularly through the Bottom Third (from ~7 persons/bedroom at the lowest ventile to ~3 by the Middle Third and Top Third border).
- The decline continues more gradually through the Middle and Top Thirds, approaching 1.5 persons/bedroom at the highest incomes.

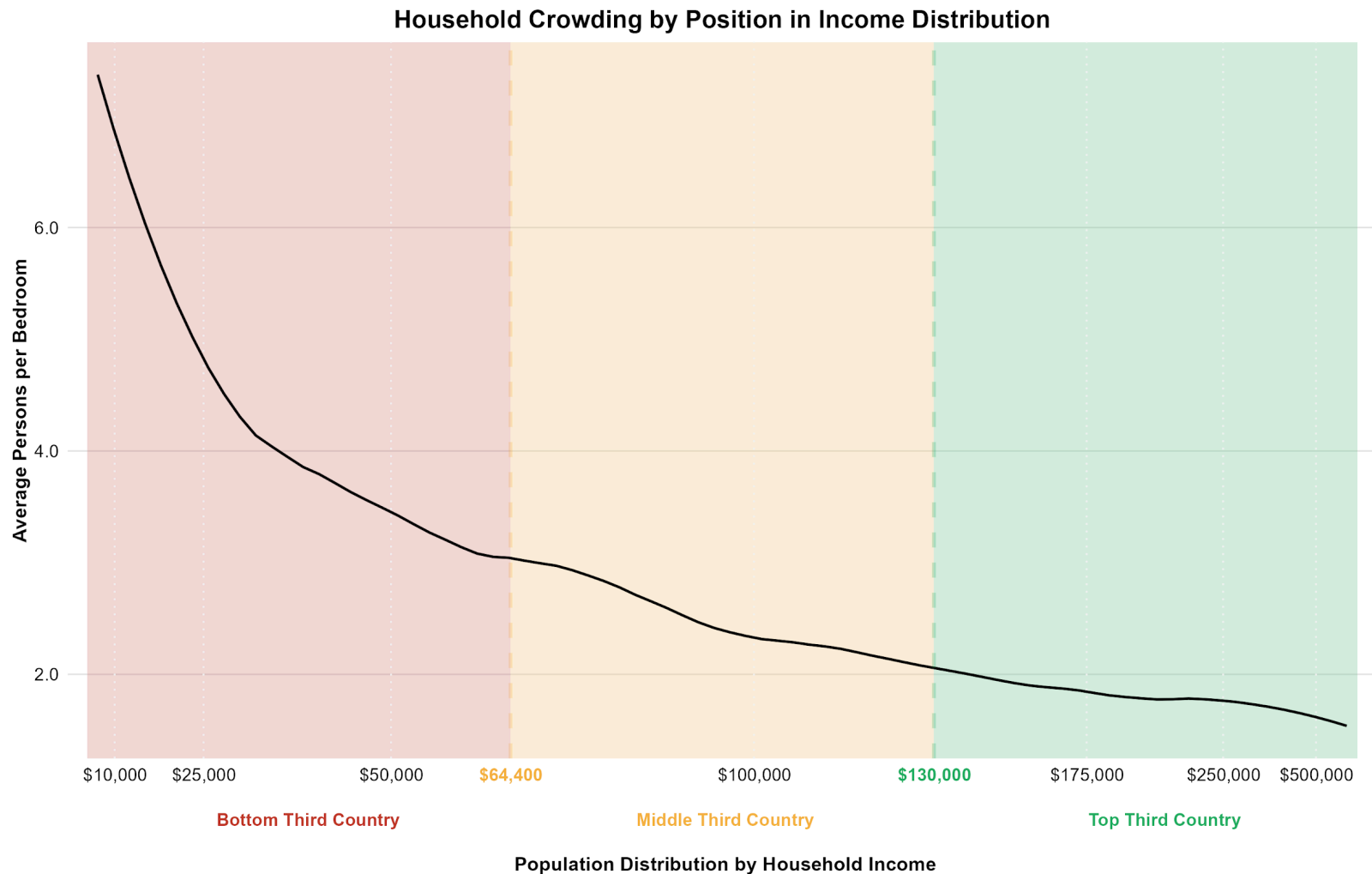


Figure 15: Technology Access

- Access to "any device" (computer, smartphone, or tablet) is relatively high across the board, but shows a slight positive trend with income.
- Access to a computer and, more significantly, broadband internet shows a much stronger positive correlation with income, rising steeply through the Bottom and Middle Thirds before plateauing at high levels (>90% for computer, >85% for broadband) in the Top Third.

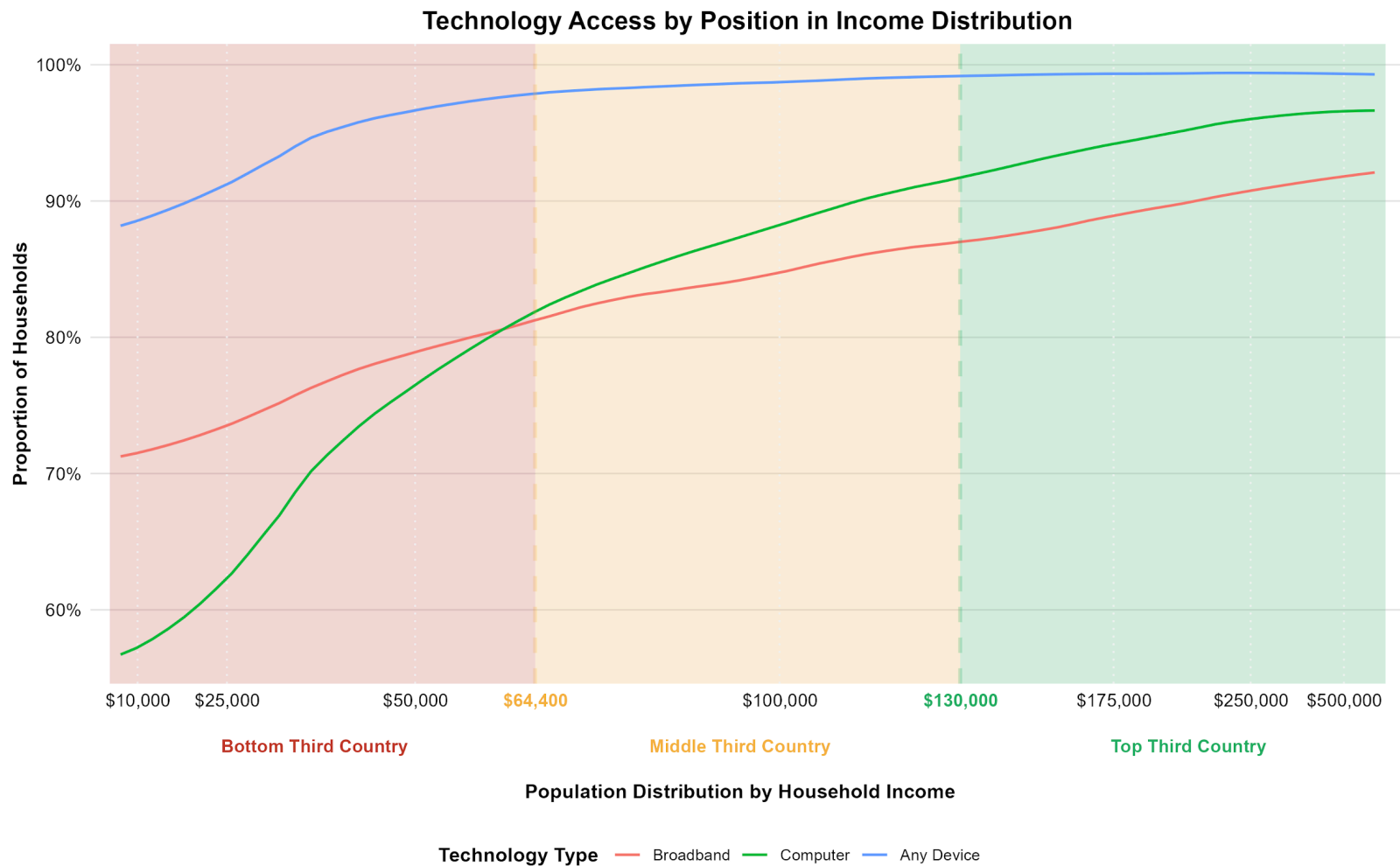


Figure 17: Self-Rated Health

- Average self-rated health score (higher score = better health) increases consistently and significantly with income position across the entire distribution, indicating a strong socioeconomic gradient in perceived health.

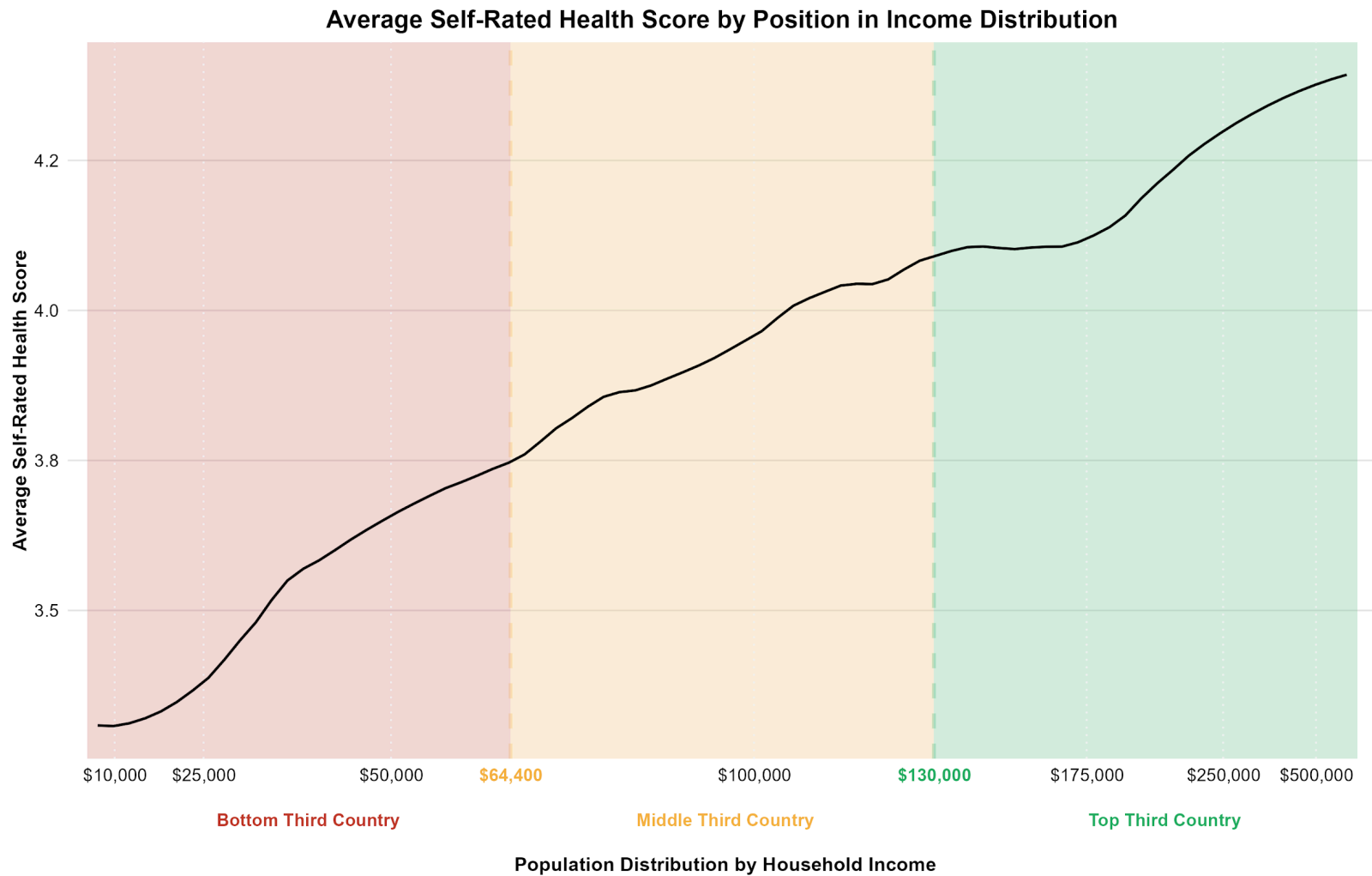


Figure 18: Disability

- The proportion reporting any difficulty decreases steadily with income. Mobility and Cognitive difficulties show the steepest declines.
- Hearing and vision difficulties also decrease, but less sharply. Self-care difficulty rates are lowest overall but also show a negative correlation with income.

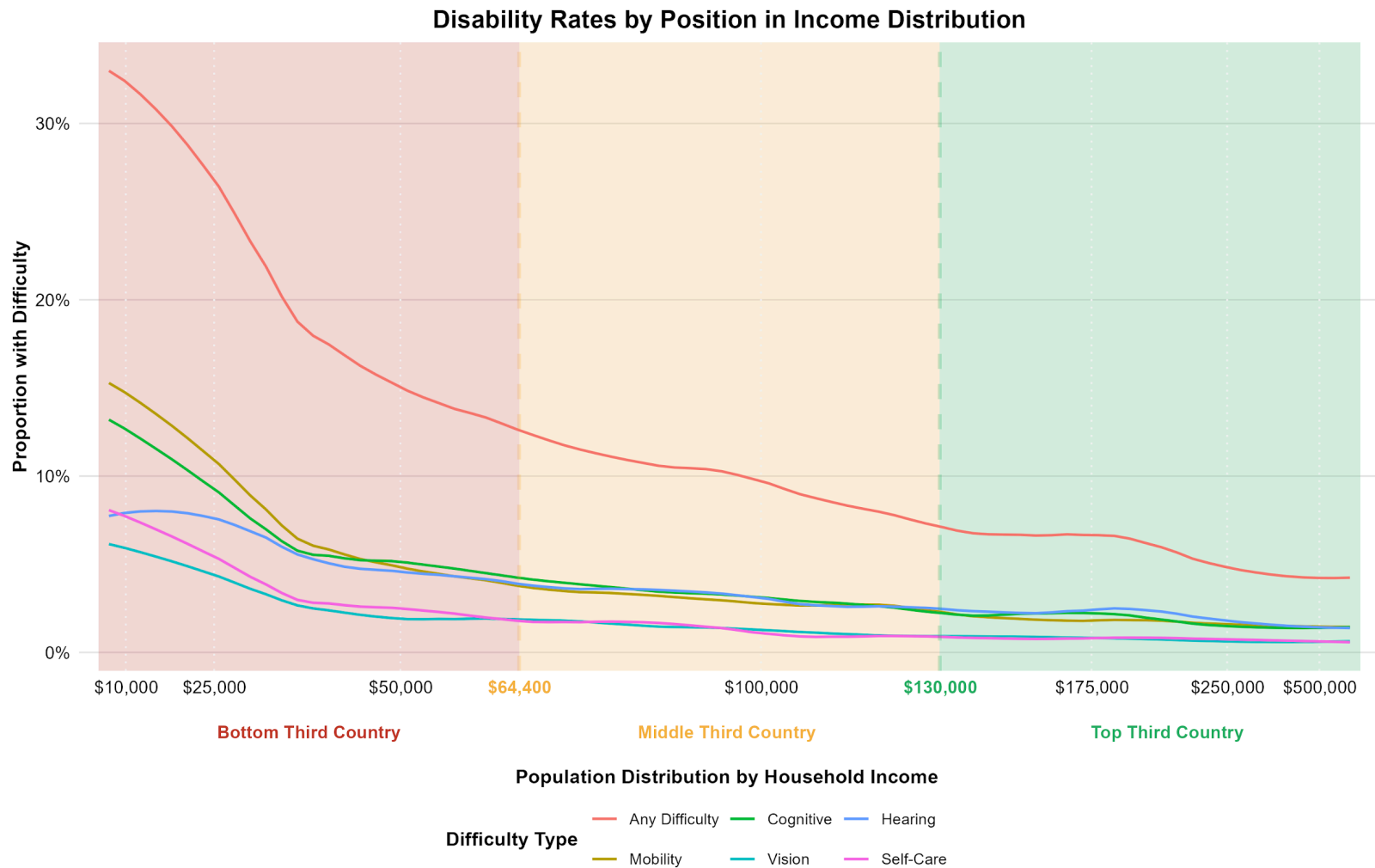


Figure 19: Heart Disease

- The reported rate of ever being told one had heart disease shows a generally negative correlation with income, decreasing through the Bottom and Middle Thirds.
- It reaches its lowest point in the lower Top Third before ticking up slightly at the very highest income levels (potentially reflecting age effects, see Fig. 1).

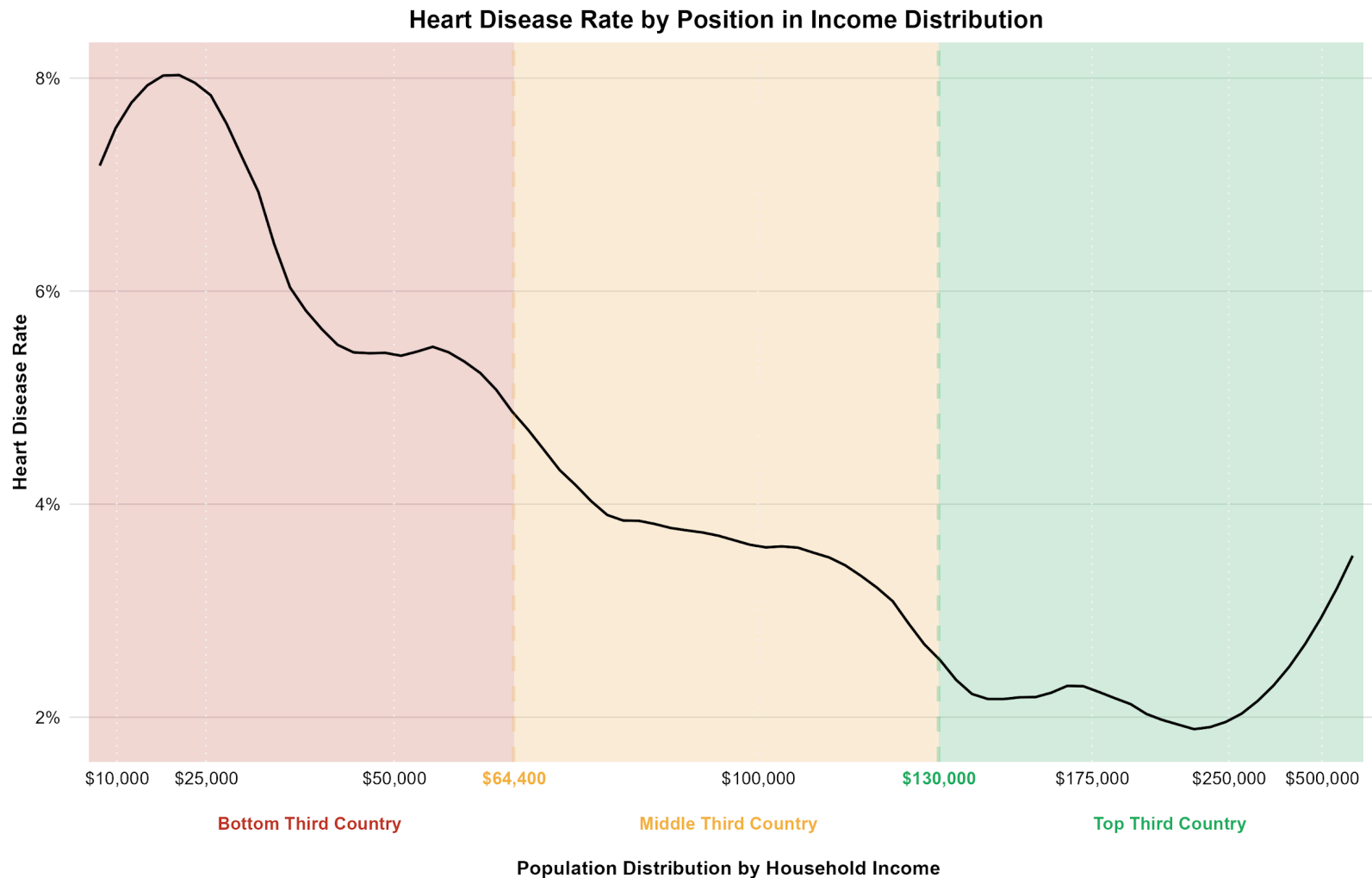


Figure 20: Obesity

- Obesity rates exhibit a complex pattern. They rise through the lower Bottom Third, peak around the \$25k-\$50k range (late Bottom Third), remain relatively high but fluctuate through the Middle Third, and then decline steadily through the Top Third.

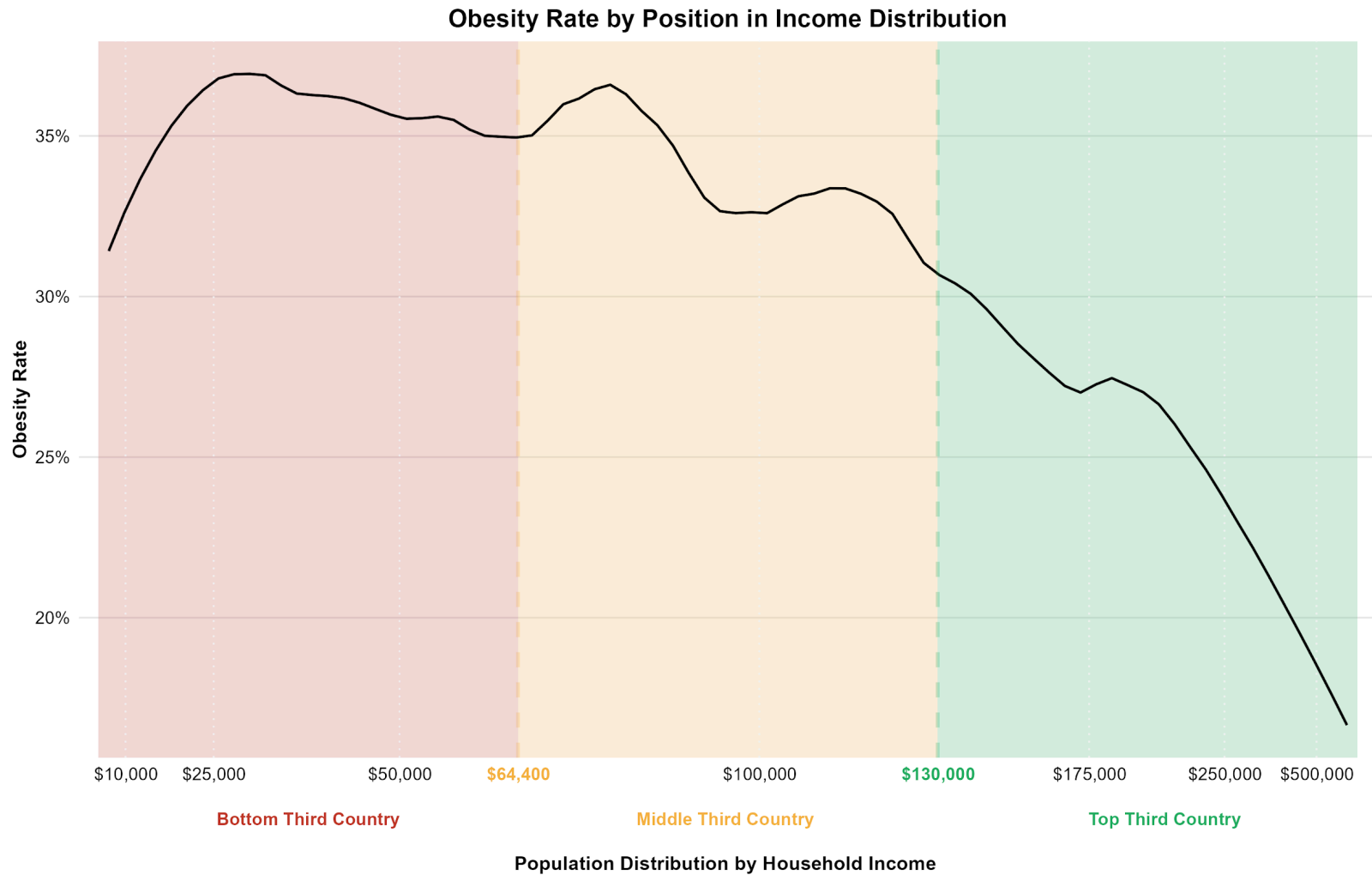


Figure 21: Smoking

- The current smoker rate (left axis) decreases significantly with income, falling sharply through the Bottom and Middle Thirds.
- Average cigarettes per day among smokers (right axis), however, shows a less clear trend, remaining relatively stable (around 10-12/day) across the Bottom Third and lower Middle Third, before peaking slightly around the Middle Third and Top Third border and then declining in the Top Third.

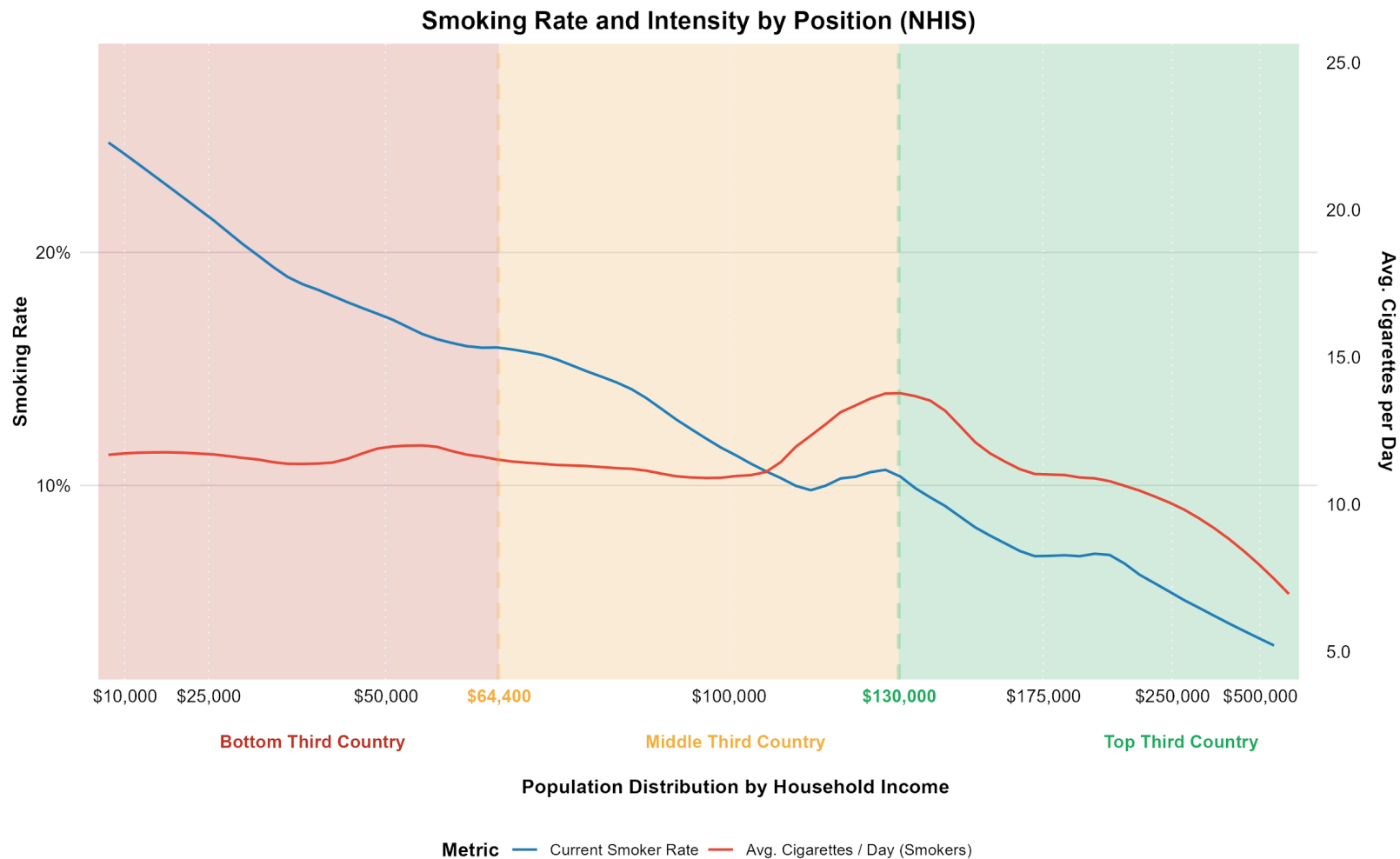


Figure 22: Non-Citizenship Rate

- The non-citizen rate is highest in the lower-middle part of the Bottom Third (peaking above 50% around \$25k-\$40k) and declines steadily as income increases through the Middle and Top Thirds, falling below 35% at the highest income levels.

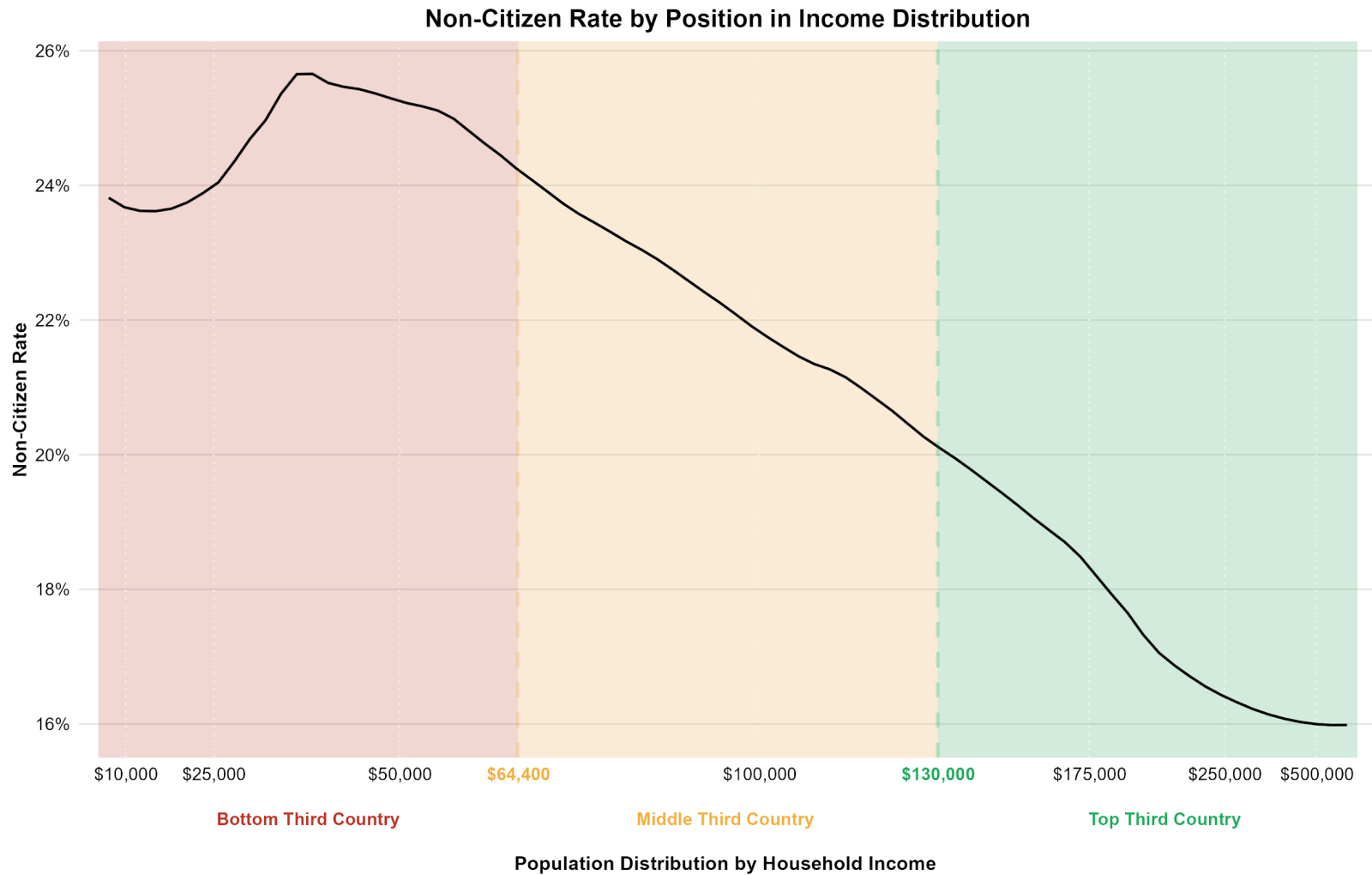
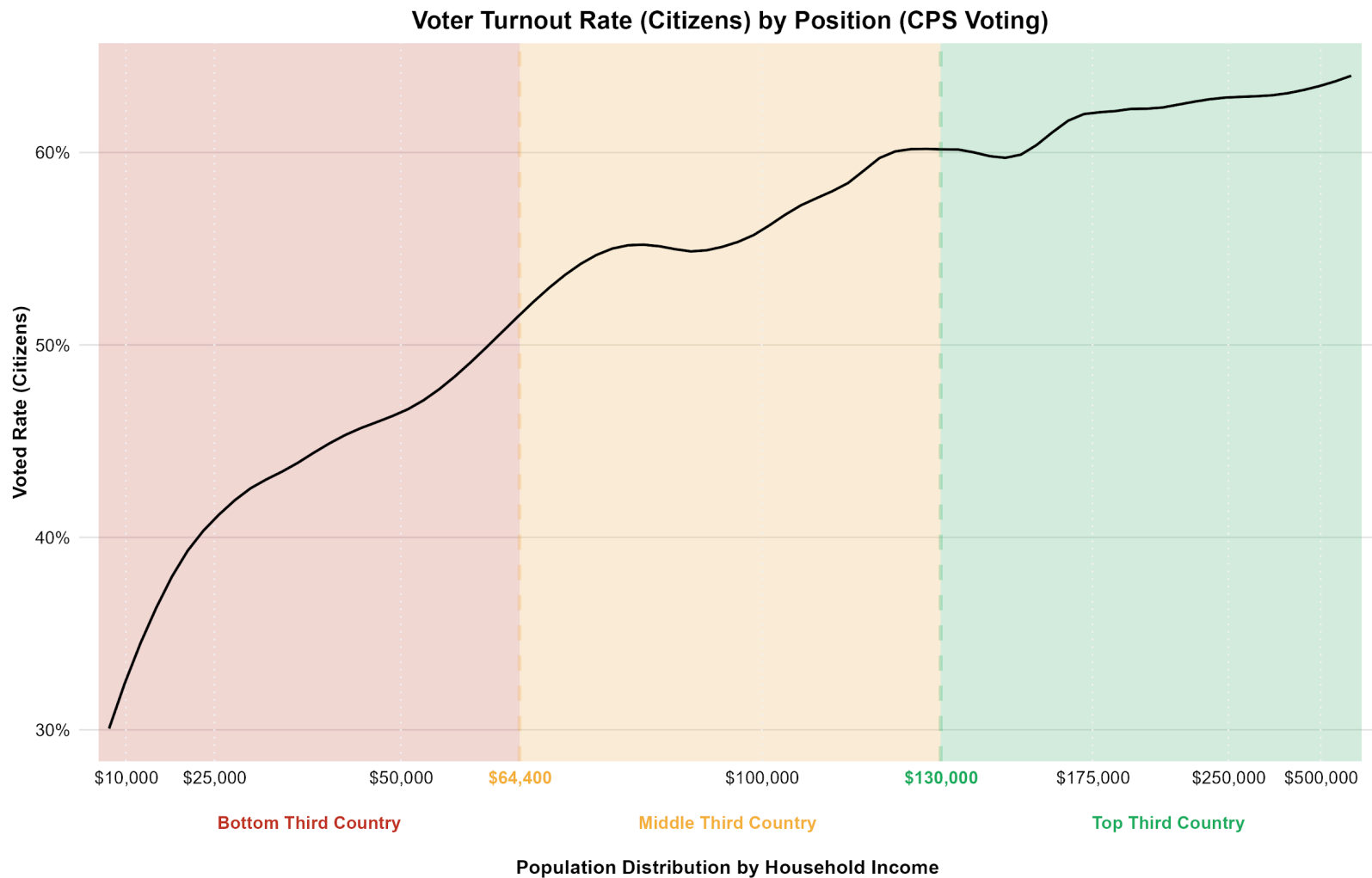


Figure 23: Voter Turnout

- Voter turnout among citizens shows a strong positive correlation with income.
- It rises steeply from around 30% at the lowest incomes through the Bottom Third, continues to increase through the Middle Third, and plateaus at high levels (over 60-65%) in the Top Third.



Foundational Readings for Income-Tier Analysis

Each main focus area—Economic Security and Opportunity, Health and Human Development, and Equity, Sustainability, and Community Resilience—is supported by a set of articles selected through a defined review process. For each indicator, a high-quality source was identified that directly connects the measure to income, poverty, or economic stratification. Selection criteria emphasized relevance, conceptual clarity, and analytical depth. Priority was given to materials that illustrate how structural inequality shapes outcomes across the income distribution. Accepted sources include academic publications, white papers, and policy reports that examine how economic and social conditions vary by income tier.

Economic Security and Opportunity

[Housing Affordability and Economic Growth](#)

Jerry Anthony

Housing Policy Debate

May, 2022

- This study investigates the relationship between housing affordability and per capita economic growth across the 100 most populous metropolitan statistical areas (MSAs) in the United States, using data from 2000, 2010, and 2015. It evaluates whether the rising share of cost-burdened households—spending more than 30% of their income on housing—has negatively impacted regional economic performance.
- Anthony argues that declining housing affordability is a pressing social concern and a structural barrier to economic development. As more households are pushed into housing cost burdens, they have less disposable income to contribute to local economies. This suppresses consumption and may alter firm behavior, deterring businesses from locating in or expanding within high-cost regions.
- The study quantifies the economic impact of diminished affordability by using publicly available data and regression models linking housing cost burden metrics to per capita GDP growth. It positions affordable housing as a welfare issue and a driver of regional competitiveness.
- **Findings**
 - Decreased housing affordability had a statistically significant negative effect on per capita GDP growth across the studied MSAs.
 - High shares of cost-burdened households were associated with slower economic expansion, suggesting that unaffordable housing undermines metropolitan prosperity.
 - The affordability crisis reduces consumer spending, strains local labor markets, and increases business costs in expensive regions.
- **Connection to 1/3s: All Thirds**
 - Bottom Third: Housing unaffordability is a growing barrier to economic stability. Low-income households spend much of their income on shelter, limiting their

ability to invest in health, education, and mobility. This perpetuates poverty and weakens local economic demand.

- Middle Third: Affordability challenges in metro areas jeopardize long-term security and homeownership, especially in high-growth markets, as rising housing costs outpace wages, putting many at risk of becoming cost-burdened.
- Top Third: Although insulated from direct housing costs, regions may face indirect effects like slower growth, labor shortages, and productivity losses due to housing inefficiencies. Ensuring affordable housing is crucial for equity and overall economic growth.

Indicator: Access to Affordable Housing

[The fading American dream: Trends in absolute income mobility since 1940](#)

Raj Chetty, David Grusky, Maximilian Hell, Nathaniel Hendren, Robert Manduca, and Jimmy Narang

American Association for the Advancement of Science

2017

- This study examines how the likelihood of U.S. children earning more than their parents—“absolute income mobility”—has changed over time. It introduces a new methodology combining census, CPS, and tax data to track generational earnings from the 1940 to 1984 birth cohorts.
- The authors argue that the American Dream is fading: the chances of children outperforming their parents economically have fallen dramatically. This decline is not primarily due to slower GDP growth, but rather to how economic growth has become increasingly unequally distributed. Broad-based growth, not just more growth, is essential to restoring upward mobility.
- They estimated absolute income mobility for each cohort by combining parent and child income distributions with a statistical tool called a “copula” (which links income ranks across generations). Using data from tax records and household surveys, they quantified the share of children earning more than their parents and tested counterfactual scenarios for growth and income distribution.
- **Findings**
 - Absolute mobility dropped from 92% for children born in 1940 to 50% for those born in 1984.
 - The largest decline occurred among middle-income families.
 - Equalizing the distribution of economic growth (as in the 1940s) would reverse 70% of the mobility decline, while increasing GDP growth alone (without redistributing gains) would reverse only about 30%.
 - Broad-based economic growth—where gains reach low- and middle-income families—is critical to restoring higher mobility levels.
- **Connection to 1/3s: Bottom Third**
 - For the Bottom Third, the erosion of broadly shared growth has severely diminished upward mobility. Children from low-income households today are far less likely to surpass their parents’ standard of living than they were in past generations. This study underscores that without addressing economic inequality and redistributing the benefits of growth, the promise of upward mobility—and with it, the American Dream—remains out of reach for the bottom third.

Indicator: Economic Mobility

Measuring Better: Development of 'True Rate of Unemployment' Data as the Basis for Social and Economic Policy

The Ludwig Institute for Shared Economic Prosperity (LISEP)

November, 2020

- The study introduces the “True Rate of Unemployment” (TRU) as a new metric to measure whether individuals are employed and whether their employment provides a living wage and full-time stability. It redefines labor market health by focusing on job quality rather than simple employment status.
- LISEP contends that the conventional BLS unemployment rate underestimates economic distress among working-age Americans by overlooking underemployment and low-wage work. The TRU seeks to realign labor metrics with actual economic experiences to better guide social and monetary policy.
- The study compares TRU and BLS figures across time and demographics, disaggregating data by race and educational attainment. It also explores trends over 25 years to assess the persistence and distribution of employment hardship.
- **Findings**
 - In January 2020, the TRU stood at 23.5%, compared to a BLS unemployment rate of just 3.6%.
 - Black Americans have consistently experienced higher levels of “true” unemployment than White Americans since at least 1995.
 - Educational attainment has a strong inverse relationship with the TRU. In October 2020, those without a high school diploma had a TRU of 50.2%, compared to 13.1% for those with advanced degrees.
 - The TRU reveals a broad and persistent gap between formal employment classification and actual economic well-being.
- **Connection to 1/3s: Bottom Third**
 - Bottom Third: The TRU exposes deep employment insecurity masked by conventional labor statistics. Many individuals in this group are formally classified as “employed” but earn insufficient wages or lack stable full-time hours. As a result, standard metrics understate the economic fragility of low-income households. These findings call for a shift in employment measurement and policy, emphasizing the need for living-wage jobs and expanded support for underemployed workers.
 - Middle Third: The TRU highlights hidden vulnerabilities in a tier often presumed stable but increasingly exposed to wage stagnation and precarious work.
 - Top Third: By contrast, it is largely unaffected by the disparities captured in the TRU, underscoring how conventional economic measures disproportionately reflect the experiences of the top third while failing to illuminate the hardship below.

Indicator: Employment

Macroeconomic Misery by Levels of Income in America

Martin Ravallion

National Bureau of Economic Research

July, 2021

- The paper investigates how core macroeconomic variables—unemployment, inflation, and GDP growth—differentially affect real household incomes across the U.S. income distribution over a nearly 30-year period (1988–2016).
- Ravallion argues that traditional macroeconomic metrics like Okun’s Misery Index (which weights unemployment and inflation equally) fail to capture how macroeconomic “misery” is unequally distributed across income levels. He contends that unemployment has a much more severe impact on low-income households than inflation, and this disparity should inform how we interpret macro indicators and set policy.
- Using data from 29 years of the U.S. Current Population Survey and national macroeconomic indicators, the study regresses household income levels across the income distribution (including a novel “income floor” measure) against unemployment, inflation, and GDP growth. It analyzes both short- and long-term effects and tests the validity of the traditional Misery Index across income groups.
- **Findings**
 - Unemployment negatively affects all income levels but disproportionately impacts the poorest and richest households, forming a U-shaped effect.
 - Inflation mainly affects middle-income groups and has minimal or inconsistent effects on the poorest and richest.
 - GDP growth supports all income levels, especially the poor, but its impact is largely through reducing unemployment.
 - Recessions raise poverty, reduce the “skewness” of income distribution (hurting the poor more than the middle), and have ambiguous effects on overall inequality.
 - The Okun Misery Index is only valid for high-income households; it significantly underrepresents hardship among low-income groups.
- **Connection to 1/3s: All thirds**
 - Bottom Third: Highly vulnerable to economic downturns; a one-percentage-point rise in unemployment increases poverty by one-third of a percentage point. Inflation has minimal impact on fundamental income changes.
 - Middle-Third: This is impacted by both inflation and unemployment, with inflation being more significant. They experience less real income loss during recessions compared to low-income individuals.
 - Top Third: Affected by unemployment through asset returns and business cycle volatility, with inconsistent inflation effects. Okun's Index is relevant but doesn't fully capture their experience.

Indicator: Income Levels

Infrastructure and the Economy

Congressional Research Service

November, 2021

- This report examines the relationship between public infrastructure investment and economic outcomes in the United States. It explores the effects of infrastructure spending on GDP, productivity, and employment, and evaluates how factors like financing method and business cycle timing affect the outcomes of public investment.
- The report argues that infrastructure investment can enhance both short- and long-term economic output, but its effectiveness depends heavily on how and when it is implemented. Deficit-financed spending tends to stimulate short-term growth and employment during downturns, while deficit-neutral investment is more favorable for long-term growth. However, public investment has declined over time, and U.S. infrastructure needs remain underfunded.
- Drawing on economic theory and macroeconomic research from the Congressional Budget Office (CBO), the Bureau of Economic Analysis (BEA), the IMF, and academic literature, the report synthesizes findings about infrastructure's macroeconomic effects. It assesses federal, state, and local investment trends and evaluates the expected impact of the Infrastructure Investment and Jobs Act of 2021.
- **Findings**
 - Well-targeted public infrastructure—especially in transportation and utilities—can increase economic output by making private sector activity more efficient.
 - Deficit-financed investments are more effective for short-term stimulus during recessions, while deficit-neutral investments are better for long-term growth without crowding out private investment.
 - Government investment has fallen as a share of GDP since the 1960s, and economists project that recent legislation (like the 2021 Infrastructure Investment and Jobs Act) will yield modest but positive economic gains if implemented effectively.
- **Connection to 1/3s: Bottom Third**

For the Bottom Third, targeted infrastructure investment—especially in transportation, broadband, and water systems—has the potential to expand opportunity, reduce underemployment, and connect underserved communities to regional economies. However, benefits depend on equitable implementation and timing. Since low-income communities are often disproportionately burdened by deteriorating infrastructure, increased federal investment could significantly improve access to jobs, services, and resilience—if these communities are prioritized in funding decisions.

Indicator: Infrastructure Quality

Metro Monitor 2024: Is Metropolitan America Benefitting from National Growth?

Glencora Haskins, Joseph Parilla, and Mayu Takeuchi

Brookings Metro

March, 2024

- This study evaluates whether the recent national economic boom has translated into inclusive growth for local communities. Drawing on 15 metrics across growth, prosperity, and inclusion, the Brookings Metro Monitor tracks economic progress from 2012 to 2022 across 194 U.S. metro areas. It examines GDP and employment growth and racial, geographic, and income-based disparities.
- The authors argue that although most metro areas experienced growth, prosperity was concentrated in the most significant regions and often failed to close racial or neighborhood gaps. While some areas—especially in the West—saw strong wage and productivity gains, others, particularly in the South, relied more on population growth than economic inclusion. In short, not all Americans have benefited equally from the post-COVID expansion.
- The report evaluates metro performance on five indicators using longitudinal data across labor market indicators, income, employment, and demographic patterns. It introduces a typology to classify metros by performance patterns and offers comparisons across regional groupings, highlighting who is benefiting and who is left behind.
- **Findings**
 - 75% of metro areas improved on all growth indicators, but prosperity gains were concentrated in large metros.
 - Metro areas with tech hubs and innovation industries (e.g., San Jose, Austin) saw the most significant prosperity increases.
 - Earnings and employment grew in nearly all regions, but racial and neighborhood income gaps remained persistent or widened.
 - 94% of metro areas improved overall inclusion, but only 40% improved on all racial inclusion indicators (employment, earnings, poverty).
 - Geographic inequality deepened: In 61% of metros, the earnings gap between the richest and poorest neighborhoods exceeded \$74,000 in 2022.
- **Connection to 1/3s: All Thirds**
 - Bottom Third: Improvements in employment and earnings haven't closed the opportunity gap. Many still live in neighborhoods left behind, where wage growth lags, and poverty remains entrenched.
 - Middle Third: Metro size and location matter. Some mid-sized cities gain ground via regional investments or tech spillovers, while others remain stagnant.
 - Top Third: Prosperity is rising fastest, especially in thriving metro cores and high-tech regions. But even here, affordability and equity challenges threaten long-term sustainability.

Indicator: Poverty

Social Mobility and Inequalities

Divya Vaid and Ankur Datta

Global Handbook of Inequality

August, 2024

- This chapter offers a sociological analysis of social mobility—inter- and intra-generational—and its links to inequality systems like caste, class, race, ethnicity, and gender. It combines macro-quantitative research with ethnographic insights and examines how mobility varies across different geographies, historical contexts, and social groups.
- The authors argue that while upward mobility is seen as a way to address rising inequality, the connection between them is complex and influenced by structural factors. They critique the focus on income-based models and call for more nuanced, class-based approaches that reflect lived experiences and intersecting identities.
- This chapter synthesizes mobility research from sociology and economics, examining historical and contemporary global trends through four "generations." It incorporates ethnographic and qualitative accounts to emphasize micro-level experiences. Key themes include the OED (origin-education-destination) triad, absolute vs. relative mobility, and comparative patterns between the Global North and South.
- **Findings**
 - Relative mobility rates in many advanced economies rose in the mid-20th century but later plateaued or declined; downward mobility is increasing.
 - While absolute upward mobility rose post-liberalization in India, relative mobility remains low and caste-bound.
 - Women's mobility remains constrained, with urban areas offering precarious opportunities and heightened risk.
 - Racial minorities in both North and South face structural barriers despite varied national trajectories.
 - Mobility is often non-linear, subject to reversals, and perceived differently across communities, with middle groups showing unpredictable trajectories.
- **Connection to 1/3s: All Thirds**
 - Bottom Third: Structural inequalities related to race, caste, or gender restrict upward mobility and often result in precarious gains or social trade-offs, such as migrant labor or leaving the workforce for status.
 - Middle Third: Vulnerable to downward mobility amid job insecurity and reduced public welfare, competition, emotional pressures, and geographic/racial disparities shape their experience.
 - Top Third: More shielded from downward mobility, this tier maintains generational status through cultural, economic, and social capital but still faces pressures from symbolic insecurity and changing labor market dynamics.

Indicator: Social Mobility

Health and Human Development

[The Economics of Child Well-Being](#)

Gabriella Conti and James J. Heckman

National Bureau of Economic Research

October, 2012

- The paper presents an integrated economic framework for understanding child development, emphasizing the importance of early childhood in shaping long-term well-being.
- Early life environments significantly influence cognitive, behavioral, and biological development, with disparities leading to lifelong inequalities.
- The authors synthesize evidence from various disciplines, including economics, biology, and social sciences, to construct a comprehensive model of child development.
- **Findings**
 - The paper documents how children from low-income households typically begin life with fewer developmental resources. These disparities can emerge as early as age 3 and become more pronounced if unaddressed.
 - The study emphasizes that early childhood is a critical period for brain and skill development. This includes both cognitive (IQ, literacy, numeracy) and non-cognitive skills (emotional regulation, motivation, social skills). These capabilities form the foundation for later success in school, the labor market, and overall life satisfaction.
 - Interventions during early childhood, especially for disadvantaged children, have the highest return on investment compared to later interventions in adolescence or adulthood. Heckman's earlier work has shown that high-quality preschool programs can deliver returns of 7–10% per year, per dollar invested, through better education, health, and earnings outcomes and reduced crime and welfare dependency.
- **Connection to 1/3s: Bottom Third**
 - The study's findings are especially relevant to low-income households, where children are more likely to face limited access to key resources like quality early education, healthcare, nutrition, and stable housing. Conti and Heckman argue that early developmental gaps in cognitive and non-cognitive skills are rooted in socioeconomic disadvantage and tend to widen over time without intervention.
 - As a result, children from low-income backgrounds are statistically more likely to enter school less prepared, face greater health challenges, and experience reduced economic mobility as adults. The study emphasizes that strategic early investments—through public policy, education, and social support—can significantly improve life outcomes, reduce intergenerational poverty, and boost national productivity.

Indicator: Child Well-being

How Cultural Factors Shape Economic Outcomes

The Future of Children

2020

- The article examines how parenting behaviors differ by socioeconomic status (SES) and how these differences contribute to gaps in children's cognitive, emotional, and academic outcomes. It emphasizes early childhood as a crucial period for development and social mobility.
- The authors argue that differences in parenting practices—especially cognitive stimulation and emotional support—are key drivers of intergenerational inequality. They contend that stress and cognitive biases, not differences in values or goals, are the main barriers preventing low-income parents from engaging in enriching behaviors.
- The study reviews decades of interdisciplinary research, including time-use surveys, longitudinal data, and randomized controlled trials, to identify patterns in parenting across income groups. It also evaluates mechanisms that explain these gaps, such as financial stress, time constraints, and psychological biases.
- **Findings**
 - High-SES parents spend more quality and educational time with children and use more responsive and less punitive discipline.
 - Parenting differences are not due to divergent values or goals—parents across SES levels want the same things for their kids.
 - Stress and cognitive overload (e.g., present bias, automaticity) impair low-income parents' ability to follow through on parenting intentions.
 - Traditional parenting interventions (e.g., home visits) often show modest impacts.
 - Light-touch behavioral interventions (e.g., goal setting via text) show promise in changing parenting behavior and boosting child outcomes.
- **Connection to 1/3s: Bottom Third**
 - For the Bottom Third, structural stress and decision-making constraints—not lack of love or effort—limit parental engagement. Programs that reduce economic stress or use behavioral tools to support parenting can help level the playing field. Targeted, low-cost interventions that respect parents' goals while addressing real-world challenges offer scalable paths toward improving intergenerational mobility.

Indicator: Cultural Enrichment

Twelve facts about the economics of education

Sofoklis Goulas

The Hamilton Project

June, 2024

- This report outlines twelve key facts that summarize the economic importance of education in the U.S., highlighting trends in student performance, educational attainment, equity gaps, the economics of higher education, and innovation efforts (e.g., EdTech, charter schools, R&D). It provides an accessible and data-rich overview of the structural and financial issues facing the education system.
- The authors argue that while education is a powerful engine for upward mobility, racial, economic, and gender-based disparities persist, exacerbated by the COVID-19 pandemic. Investments in innovation, equity-focused reforms, and evidence-based policy are needed to enhance education's role in fostering inclusive prosperity.
- Using federal data sources (e.g., NCES, PISA, CPS, BLS), the report presents trends in educational performance, attainment, costs, and returns, comparing these across race, gender, income, and over time. It highlights disparities in access, outcomes, and affordability, while evaluating innovation strategies like charter schools, EdTech, and education R&D investments.
- **Findings**
 - Despite gains in high school graduation and college enrollment, Black and Hispanic students remain significantly less likely to complete postsecondary education and face greater unmet financial need, limiting upward mobility.
 - Earnings have grown more for those with higher education, especially women, but gender and racial pay gaps persist across all levels of attainment. Men still earn more than women with the same degree, and access alone is not closing these gaps.
 - Charter schools show promise, particularly for marginalized communities, but results vary. Investment in EdTech surged during COVID-19 and then collapsed, and federal funding for education R&D has declined, underscoring the urgent need for sustained, evidence-based innovation.
- **Connection to 1/3s: Bottom Third**
 - For the Bottom Third, this study underscores both promise and peril. While high school graduation rates are improving, low-income students—especially those who are Black or Hispanic—face enduring barriers to college access, affordability, and completion. The post-COVID education system requires targeted innovation and funding to close opportunity gaps. Without deeper public investment in proven, inclusive educational models, the benefits of education will continue to disproportionately flow to higher-income groups.

Indicator: Educational Attainment

[Access to Healthcare and Disparities in Access](#)

Agency for Healthcare Research and Quality (AHRQ)

U.S. Department of Health and Human Services

December, 2023

- Mental health needs—especially among adolescents—have grown significantly, but access to treatment remains limited. Suicide rates among teens rose by over 75% since 2008, and fewer than half of those with depression received care, with even lower treatment rates for low-income and Hispanic youth.
- Long-stay nursing home residents experienced worsening physical health during the pandemic, with increased reports of pain, mobility loss, and reduced access to daily care. Staffing shortages and isolation measures compounded these declines in wellness.
- Telehealth expanded rapidly during COVID-19, particularly for behavioral health services, but access was uneven. Older adults, rural residents, and low-income individuals were far less likely to use or benefit from these services due to technology and access barriers.
- **Findings**
 - The report provides a comprehensive overview of healthcare delivery and health outcomes across the U.S. It highlights persistent disparities in access, quality, and population wellness, especially during COVID-19.
 - The report emphasizes how health systems responded to the pandemic, focusing on both mental and physical well-being. It explores how infrastructure gaps, socioeconomic stress, and systemic inequities shaped health outcomes across geography, race, and income.
 - The authors argue that traditional health system performance metrics are too narrow. They call for greater inclusion of mental health, functional health, and behavioral healthcare access, especially for marginalized groups.
- **Connection to 1/3s: All Thirds**
 - Bottom Third: These populations were disproportionately affected by rising suicide rates, untreated depression, delayed care, and limited access to telehealth. Chronic understaffing in long-term care facilities and inadequate behavioral health support amplified mental distress.
 - Middle Third: Gaps in healthcare access and rising out-of-pocket costs challenged both mental resilience and physical wellness, while deferred preventive care created long-term health risks.
 - Top Third: This group remained more buffered, benefiting from access to both traditional and telehealth services. Yet, even among this group, mental health strain rose during the pandemic, highlighting that while the burden is greatest for Low-Income, mental and physical wellness is now a national crisis.

Indicator: Healthcare Access and Quality

[Ten Americas: A Systematic Analysis of Life Expectancy Disparities in the USA](#)

Laura Dwyer-Lindgren et al.

The Lancet

December, 2024

- This study expands the Eight Americas framework by defining ten distinct population groups to analyze life expectancy disparities in the U.S. from 2000 to 2021. The authors aim to show how long Americans live and who has seen improvements or declines in life expectancy, emphasizing the roles of race, income, geography, and segregation.
- Dwyer-Lindgren et al. highlight that national averages mask significant inequities among subgroups worsened by COVID-19. They argue that longevity disparities result from complex interactions of systemic inequalities rather than solely from income or race.
- Using data from the National Vital Statistics System and the U.S. Census Bureau, the study calculates life expectancy for each group based on race/ethnicity, income, metropolitan status, and segregation. It also examines changes over time by age and sex and correlations to education and income levels.
- **Findings**
 - Life Expectancy gaps are widening. In 2021, the gap between the longest- and shortest-living groups reached 20.4 years, with American Indian/Alaska Native populations in the West (America 10) facing persistent declines even before COVID-19.
 - There were uneven impacts of the pandemic. Life expectancy dropped across all groups in 2020, with only minimal rebounds in 2021. Latino and Black communities saw especially sharp setbacks after earlier progress.
 - There are structural and systemic drivers. While income and education generally correlate with longevity, anomalies, like high-income America 3 having only moderate life expectancy, point to deeper systemic and structural inequalities affecting health outcomes.
- **Connection to 1/3s: All Thirds**
 - Bottom Third: Factors like location, race, and segregation greatly impact life expectancy beyond income. Groups such as American Indians and Alaskan Natives in the West, Black Americans in the Deep South, and Whites in Appalachia face declining life expectancy due to issues like poor healthcare access and structural racism. Progress in many Black communities has stalled since 2010, and COVID-19 has further increased mortality risks.
 - Middle Third: In predominantly white suburban areas, high income and education have not led to improved health outcomes, highlighting vulnerability to health shocks and an increase in midlife mortality.
 - Top Third: Asian Americans enjoy a longer life expectancy due to stability, experiencing minimal impact from COVID-19.

Indicator: Life Expectancy

[Making lifelong learning a reality: a handbook](#)

UNESCO Institute for Lifelong Learning

2022

- This study outlines twelve key facts that reveal how education shapes individual and national economic well-being in the U.S. It covers secondary and postsecondary education trends, disparities in outcomes, the economics of higher education, and areas of innovation like charter schools, EdTech, and R&D investment.
- The report argues that education remains a cornerstone of upward mobility, but progress is uneven. Persistent racial and economic disparities, worsened by COVID-19, limit equitable access to opportunity. Addressing these challenges requires sustained public investment, innovation, and policy reform grounded in evidence.
- The authors synthesized national datasets (e.g., NCES, CPS, PISA, BLS) to present trends in academic achievement, educational attainment, college affordability, labor market returns, and educational innovation. They analyzed outcomes across racial, gender, and income groups and compared national and international performance.
- **Findings**
 - Student achievement has plateaued, with setbacks from the COVID-19 pandemic and persistent racial gaps in proficiency and attainment.
 - Racial disparities in college access and financial burden remain large. Black and Hispanic students are significantly more likely to face unmet financial need.
 - The return to education is growing, but men continue to earn more than women at every level.
 - Charter schools and online learning have expanded, offering promise but with uneven performance.
 - Investment in EdTech surged during the pandemic but has since declined; federal funding for education R&D has dropped nearly 29% in real terms since 2016.
- **Connection to 1/3s: Bottom Third**
 - For the Bottom Third, this study shows that while education remains a key path to mobility, systemic inequities in access, affordability, and outcomes persist. Rising educational attainment alone is not enough—targeted investments in K–12 quality, affordable college, digital equity, and proven innovations like high-performing charter schools and EdTech are critical to close opportunity gaps and deliver on the promise of education-driven prosperity.

Indicator: Lifelong Learning

2023 National Healthcare Quality and Disparities Report

Agency for Healthcare Research and Quality (AHRQ)

U.S. Department of Health and Human Services

December, 2023

- This report reviews healthcare quality, access, and disparities in the U.S. as of 2019, covering over 250 indicators in areas like person-centered care, patient safety, care coordination, and disparities by income, race, geography, and insurance status.
- The report highlights improvements in national healthcare quality but notes persistent disparities among income and racial groups. It emphasizes that structural barriers in affordability, access, and provider diversity hinder healthcare equity, especially for low-income and marginalized populations.
- The study compares healthcare experiences and outcomes across racial/ethnic groups, income brackets, insurance types, and locations using federal partners and national surveys. It focuses on trends in access, quality, disparities, care gaps, and emerging health threats like opioids and mental health.
- **Findings**
 - There were persistent disparities in care quality and access. Low-income, uninsured, and rural populations receive lower-quality care across most metrics, with little improvement over time, especially in areas like dental and routine health access.
 - The study revealed racial and geographic inequities. Black, Hispanic, and rural Americans face worse health outcomes and limited access to specialized care, while urban communities deal with distinct challenges such as higher rates of HIV and asthma.
 - There were systemic barriers in the health workforce. Shortages, especially in underserved areas, and a lack of diversity in leadership and high-paying specialties undermine care quality, outcomes, and patient trust.
- **Connection to 1/3s: All Thirds**
 - Bottom Third: Faces significant health challenges due to insurance gaps, high costs, and limited access to care, resulting in worse health outcomes and greater reliance on emergency services. Targeted interventions are essential for these households.
 - Middle Third: Experiences variable care quality and rising out-of-pocket costs, making them vulnerable without affordability reforms.
 - Top Third: Benefits from better access and quality of care, stable insurance, and well-served areas, but their experiences can obscure broader disparities faced by other income groups.

Indicator: Mental and Physical Wellness

Still Left Behind: How America's Schools Keep Failing Our Children

Katharine B. Stevens and Meredith Tracy

American Enterprise Institute

2020

- This report examines how well U.S. public schools serve lower-income students by analyzing national trends in 8th-grade reading and math achievement from 2003 to 2017. Using NAEP data, the authors focus on the percentage of students scoring below the “Basic” level—those failing to demonstrate even minimal competence.
- Despite decades of school reform and significantly increased education spending, large proportions of lower-income students still fail to meet basic benchmarks in reading and math. The report argues that persistent achievement gaps are obscured by overreliance on average scores, and that superficial gains often mask continuing systemic failure for disadvantaged students.
 - The authors analyzed national NAEP data for 8th-grade students, comparing performance in 2003 and 2017. They focused on students eligible for free or reduced-price lunch (FRL) as a proxy for low income and measured the percentage scoring below the “Basic” level in both subjects. They also calculated gaps between lower- and higher-income students and examined how the FRL group’s composition changed over time.
- **Findings**
 - In 2017, 36% of lower-income 8th graders scored below NAEP Basic in reading and 45% in math, indicating they lacked minimum proficiency.
 - Achievement gaps between lower- and higher-income students remained wide and, in many cases, either worsened or stayed the same.
 - The FRL group became more socioeconomically mixed over time, potentially inflating perceived performance gains.
 - Increased education spending did not translate into commensurate improvements in basic student achievement.
- **Connection to 1/3s: Bottom Third**
 - For the bottom third, the findings underscore a national failure to deliver on the promise of public education as a vehicle for mobility. Despite significant financial investments and reform efforts, a large share of low-income students remain unable to demonstrate even basic literacy and numeracy after nearly a decade in school. This calls for a reevaluation of educational strategies and a greater focus on actual, meaningful outcomes, especially for the most vulnerable students.

Indicator: Quality of Education

Equity, Sustainability, and Community Relations

[Beyond Connectivity: The Role of Broadband in Rural Economic Growth and Resilience](#)

Adam Dewbury, May Erouart, and Dr. Amanda Weinstein

Center for Rural Innovation

2024

- This report investigates how broadband access and utilization affect economic growth and resilience in rural U.S. communities. It analyzes the relationship between broadband infrastructure, broadband adoption, and various indicators of local economic dynamism, such as business growth, income, and employment.
- The authors argue that broadband is not just basic infrastructure—it is a transformative tool for rural prosperity when effectively utilized, especially in areas served by small, community-based providers. High broadband utilization (measured by adoption rates and provider engagement) can reverse rural decline, drive entrepreneurship, and reduce underemployment.
- The team used a propensity score matching methodology to compare rural counties with high versus low broadband utilization. They controlled for income, education, race, and urbanization to estimate the causal impact of broadband on local outcomes. Key indicators included business growth, self-employment, per capita income, and GDP.
- **Findings**
 - Compared to similar counties with lower utilization, rural counties with high broadband utilization had:
 - 213% higher business growth (firm growth rate)
 - 10% higher self-employment growth
 - 18% higher per capita income growth
 - 44% higher GDP growth
 - Low broadband utilization was associated with business closures, population decline, and weaker income and GDP growth.
- **Connection to 1/3s: Bottom Third**
 - For the Bottom Third, especially in rural areas, broadband access alone is not enough. Utilization matters—broadband must be affordable, high-speed, and embedded in local economic ecosystems (e.g., through incubators, telehealth, public WiFi, and digital literacy programs). When done right, broadband infrastructure helps create inclusive entrepreneurial ecosystems that support local business growth, boost incomes, and reduce underemployment—key drivers of upward mobility for low-income households.

Indicator: Access to Technology

Food Deserts and the Causes of Nutritional Inequality

Hunt Allcott, Rebecca Diamond, Jean-Pierre Dubé, Jessie Handbury, Ilya Rahkovsky, and Molly Schnell

National Bureau of Economic Research

November, 2018

- The paper investigates why wealthier Americans eat more healthfully than poorer Americans, focusing on whether unequal access to healthy food, commonly referred to as “food deserts,” is a significant driver of nutritional inequality in the United States.
- The authors argue that nutritional inequality is primarily driven by demand-side factors—differences in preferences, knowledge, and education—rather than supply-side issues like access to healthy food. They challenge the prevailing belief that improving food access (e.g., adding supermarkets) will meaningfully reduce unhealthy eating among low-income households.
- The study used a combination of large-scale grocery purchase data from Nielsen, supermarket entry and location data, and surveys on health knowledge and preferences. Structural demand modeling with price instruments, to analyze how much of the nutrition-income gap is attributable to food access versus preferences. They conducted event studies on supermarket entry, household migration analysis, and within-location comparisons, then simulated counterfactual policies using a demand model.
- **Findings**
 - Only 9% of the nutrition-income gap is explained by differences in access to healthy foods; 91% is due to differences in demand.
 - Supermarket entry and moving to healthier areas had statistically small or insignificant effects on dietary quality.
 - Even among people shopping at the same stores, significant nutritional inequality remains.
 - Education and nutrition knowledge are strong predictors of healthy food demand.
 - A means-tested healthy food subsidy could close the nutrition-income gap for a cost equal to about 15% of the SNAP budget.
- **Connection to 1/3s: Bottom Third**
 - Nutritional inequality is not primarily a matter of access but of preferences, knowledge, and education. Simply increasing the availability of healthy foods will do little to close the diet gap. Policies like subsidies for healthy food purchases, nutrition education, and interventions that shift food preferences would be more impactful in promoting health and reducing inequality in low-income households. This insight supports a broader focus on human development and behavioral change, not just physical infrastructure, when designing equitable public health and food policies.

Indicator: Access to Water and Food

Geographies of Insecure Water Access and the Housing–Water Nexus in US Cities

Katie Meehan, Jason R. Jurjevich, Nicholas M. J. W. Chun, and Justin Sherrill

Proceedings of the National Academy of Sciences

November 2020

- This study analyzes urban water insecurity in the 50 largest U.S. metropolitan areas, exploring who lacks piped water access and the systemic socioeconomic and housing factors contributing to "plumbing poverty." It introduces the "housing–water nexus" as a framework to understand structural inequalities in urban water access.
- Meehan and co-authors contend that gaps in water access stem from deep-seated racial and economic disparities, particularly in housing, rather than just infrastructure issues. They challenge the idea that water insecurity is solely due to rural isolation or municipal failure, demonstrating that many urban residents lack basic water access because of housing instability, racial inequality, and income segregation.
- Using microdata from the American Community Survey (2013–2017) and multivariate logistic regression, the authors analyze the sociodemographic, geographic, and housing characteristics linked to lack of piped water, focusing on over 220,000 urban households facing water insecurity.
- **Findings**
 - Over 1.1 million Americans lack piped water; 73% live in urban areas.
 - Households headed by people of color are nearly 35% more likely to lack piped water than white, non-Hispanic households.
 - Renter-occupied households are 1.61 times more likely, and mobile home residents are 1.89 times more likely to lack piped water access than owner-occupied and non-mobile homes.
 - Plumbing poverty is strongly associated with low household income and occurs most frequently in cities with high-income inequality, such as San Francisco, Portland, and New York.
 - The Gini coefficient is a significant predictor: each 10% rise in metro-area inequality raises the odds of a household lacking piped water by 49%.
 - Despite assumptions, racial segregation and cost-burdened status were not statistically significant predictors once tenure and income were accounted for.
- **Connection to 1/3s: All Thirds**
 - Bottom Third: These households, especially renters and people of color, face plumbing poverty, revealing systemic housing inequities that affect health and dignity. While plumbing poverty is less common among middle-income households, it remains a risk amid rising housing costs and failing infrastructure.
 - Top Third: Households enjoy better water security, but benefit from a system that often overlooks marginalized communities.

Indicator: Access to Water and Food

How socio-economic inequality affects individuals' civic engagement: a systematic literature review of empirical findings and theoretical explanations

Joris Melchior Schroder and Michaela Neumayr

Socio-Economic Review

2023

- This systematic literature review explores how socio-economic inequality affects individuals' civic engagement, specifically in the forms of charitable giving, volunteering, and non-profit membership. It examines both empirical findings (from 70 studies) and theoretical explanations for the relationship between inequality and civic participation.
- The authors argue that rising inequality generally undermines civic engagement by eroding social cohesion, reducing trust, creating resource and power disparities, and discouraging participation among less affluent individuals. While some theories suggest inequality might motivate activism or altruism, the dominant patterns point toward reduced engagement in unequal societies.
- Using a systematic literature review (SLR) approach, the authors analyzed 70 studies across sociology, political science, economics, and psychology. They synthesized findings on the incidence and intensity of civic engagement with inequality and categorized theoretical mechanisms into five conceptual frameworks.
- **Findings**
 - Most empirical evidence shows a negative relationship between socio-economic inequality and civic engagement, particularly for charitable giving and non-profit membership.
 - The relationship is moderated by individual factors like income, education, age, and race/ethnicity.
 - The authors identify five main mechanisms that explain the effects of inequality. Social Disintegration Hypothesis – inequality reduces trust and social cohesion. Relative Power Hypothesis – power imbalances demotivate participation. Resource Hypothesis – unequal access to civic participation resources. Conflict Hypothesis – inequality can spur activism and engagement through perceived injustice. Inequality Aversion Hypothesis – altruistic concern for inequality may boost giving.
- **Connection to 1/3s: Bottom Third**
 - For the Bottom Third, the findings suggest that socio-economic inequality not only reduces access to civic life, but also weakens social trust, reduces perceived efficacy, and limits participation in institutions that traditionally serve as vehicles for voice and influence. Low-income communities may disengage due to disillusionment, lack of time or money, or feeling politically and socially powerless. As a result, public goods supported by civic engagement may deteriorate further, creating a feedback loop that deepens disadvantage.

Indicator: Civic Engagement

Climate Change and Social Vulnerability in the United States

Environmental Protection Agency

September, 2021

- This report analyzes how six major climate-related impacts—air quality, extreme temperature (health and labor), coastal flooding (traffic and property), and inland flooding—affect socially vulnerable populations in the U.S. It examines whether low-income individuals, people of color, those without a high school diploma, and adults 65+ are disproportionately at risk from climate hazards under future warming and sea level rise.
- The authors argue that socially vulnerable populations are more likely to live in areas where the most severe climate impacts will occur. As such, climate change not only amplifies environmental hazards but also reinforces existing social and economic inequalities unless equitable adaptation strategies are implemented.
- The EPA used a four-step modeling approach: (1) project climate hazards across the contiguous U.S., (2) estimate human health and economic impacts, (3) identify areas with the highest impacts, and (4) calculate how likely socially vulnerable groups are to live in these areas relative to their reference populations. Analyses were performed for scenarios of 2°C global warming and 50 cm sea level rise, among others.
- **Findings**
 - Minorities face the highest overall climate risks: Black individuals are 40% more likely to live in areas with the highest temperature-related mortality; Hispanic and Latino individuals are 43% more likely to live in areas with the greatest projected labor losses.
 - Low-income individuals and those without a high school diploma are about 25% more likely to live in areas with the highest labor losses, and 15% more likely to live in areas facing the worst air quality and sea level rise impacts.
 - Children of color face especially high risks of asthma from air pollution. Black children are 41% more likely to live in high-risk areas.
 - People 65 and older generally do not face significantly higher climate impact risk compared to younger populations, though specific regional risks exist.
- **Connection to 1/3s: Bottom Third**
 - For the Bottom Third, climate change poses a compounding threat to already limited economic and health resilience. Bottom Third populations are more likely to face harmful air quality, labor disruptions from heat, flooding of homes and infrastructure, and reduced access to adaptation benefits (like protective infrastructure). These overlapping exposures intensify vulnerability and threaten long-term opportunity and well-being, especially without intentional, equity-centered climate adaptation policies.

Indicator: Climate Change Resilience

Inclusive Innovation for U.S. Economic Growth and Resiliency

Tom Guevara, Alexander Kersten, and Srishti Khemka

Center for Strategic and International Studies

May, 2023

- This report explores how inclusive innovation—broadening participation in entrepreneurship, STEM fields, and patenting—can drive U.S. economic growth, resilience, and regional equity. It highlights structural barriers that limit underrepresented groups' access to innovation ecosystems and examines strategies to promote greater inclusion across race, gender, geography, and socioeconomic status.
- The authors argue that increasing inclusion in innovation is both an economic necessity and a moral imperative. Failing to tap into the full diversity of U.S. talent suppresses economic output, limits productivity growth, and reinforces regional and demographic inequality. Inclusive innovation improves idea generation, commercialization, and long-term national competitiveness.
- Drawing on expert insights from a CSIS-hosted panel, the report synthesizes current data on patenting, STEM education, entrepreneurship trends, and federal DEI initiatives. It evaluates barriers to entry in the innovation and patent systems and discusses federal policy responses.
- **Findings**
 - Racial and gender gaps in innovation cost the U.S. trillions in unrealized GDP. Inclusive innovation could increase U.S. GDP by up to \$640 billion.
 - Structural barriers persist in patenting, STEM education, and venture capital funding. The innovation economy is dominated by white, male, and urban actors.
 - Place-based programs like regional innovation hubs and manufacturing extension partnerships (MEPs) are promising but must be coupled with people-based measures to ensure benefits reach underserved individuals.
 - The U.S. patent system and legal profession lack diversity; patent law eligibility and STEM education barriers restrict broader participation in innovation.
 - Federal action is growing, but more robust, data-informed inclusion strategies are needed.
- **Connection to 1/3s: Bottom Third**
 - For the Bottom Third, inclusive innovation presents a transformative economic opportunity. Low-income communities are often disconnected from the innovation economy by geography, education, and institutional access. Federal investments in regional hubs, STEM education, entrepreneurship training, and patent law diversity can help unlock new pathways to prosperity. However, to succeed, policies must prioritize both place and people, ensuring that resources flow not only into underinvested regions but also to historically excluded individuals within those communities.

Indicator: Community Inclusiveness

Understanding the Relationship Between Intergenerational Mobility and Community Violence

Mann, Edin, and Shaefer

Proceedings of the National Academy of Sciences

August, 2024

- This study investigates whether low intergenerational economic mobility at the community level predicts higher levels of violence, specifically violent crime and homicide, across U.S. counties. While prior research has shown that violence impairs upward mobility, this study explores the reverse: whether blocked opportunity itself may generate violence.
- The authors argue that depressed intergenerational mobility is a structural condition that fosters frustration, weakens community institutions, and contributes to cycles of violence. They propose a feedback loop: not only does violence inhibit mobility, but low mobility also fosters conditions that lead to more violence.
- Using Raj Chetty and Nathaniel Hendren's county-level data on intergenerational mobility for low-income children (born 1980–1986), the authors modeled how mobility scores predicted violent crime and homicide rates in 2008, when the cohort reached early adulthood. They controlled for standard predictors of violence (poverty, inequality, unemployment, law enforcement presence), past violence, and county characteristics (urbanicity, housing stability, outmigration, etc.).
- **Findings**
 - Counties with lower intergenerational mobility had significantly higher rates of violent crime and homicide, even after controlling for traditional predictors and past violence.
 - Intergenerational mobility was a more consistent and robust predictor of violence than poverty, inequality, or law enforcement levels.
 - The relationship held across rural and urban counties and remained statistically significant even after accounting for outmigration and state fixed effects.
 - A 10-point drop in county mobility score (from the 46th to 36th percentile) was associated with a 91% increase in homicide and a 107% jump in violent crime.
- **Connection to 1/3s: Bottom Third**
 - For the Bottom Third, this study powerfully underscores that lack of economic mobility is not just an economic issue—it is a public safety issue. When communities are cut off from pathways to advancement, they may experience higher levels of frustration, social disorganization, and ultimately violence. Interventions aiming to reduce crime must pair policing with real, sustained investments in mobility, including education, job access, and neighborhood revitalization. This reinforces that mobility is foundational to resilience, safety, and long-term community well-being.

Indicator: Community Safety

Examining Contextual and Nonlinear Associations Between the Living Environment and Life Satisfaction

Xinyi Wu

University of Minnesota

2021

- This dissertation explores how transportation infrastructure and neighborhood design influence subjective life satisfaction across income levels in United States. Using survey data from the Twin Cities, the study examines whether the assumed linear benefits of neighborhood features apply to residents with varying socioeconomic backgrounds.
- Wu argues that urban planning often ignores how infrastructure benefits vary by income. Lower-income communities value basic improvements more than higher-income ones, who prioritize aesthetics. This suggests the need for more equitable and context-sensitive planning.
- The study analyzes data from 360 households in six Twin Cities neighborhoods using gradient boosting decision trees (GBDT) and impact asymmetry analysis (IAA) to evaluate the marginal utility of 31 neighborhood features. It compares outcomes in low- and high-income areas and also includes cross-national comparisons with urban data from Guangzhou, China.
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- **Findings**
 - In low-income neighborhoods, features related to environmental safety and basic infrastructure (e.g., air quality, noise, drainage) influence life satisfaction more than in higher-income areas.
 - In high-income neighborhoods, satisfaction is more strongly associated with cognitive and aesthetic elements (e.g., bike trails and school proximity).
 - Transportation infrastructure (sidewalks, trails) demonstrates nonlinear effects—either diminishing returns beyond a certain threshold or positive impact only after reaching basic adequacy.
 - Neighborhood satisfaction depends more on physical conditions, while life satisfaction is more tied to social connectedness and leisure access.
- **Connection to 1/3s: All Thirds**
 - Low-Income America: Life satisfaction is greatly influenced by investments in basic neighborhood infrastructure like drainage, safety, and sidewalks, which enhance physical health, mobility, and community integration.
 - Middle-Income America: This group benefits from foundational and aspirational features, requiring balanced investments for reliability and upward mobility.
 - High-Income America: These individuals derive less satisfaction from infrastructure, with gains coming more from enhancements than necessities. Wu's findings highlight the need for income-sensitive urban planning that prioritizes equity in neighborhood design and services.

Indicator: Living Environment

Income Inequality and Policy Responsiveness

Robert S. Erikson

Department of Political Science, Columbia University

March, 2015

- The study examines the relationship between economic inequality and political responsiveness in the U.S. to see how income disparities influence policy outcomes.
- Erikson argues that while low-income voters have less political influence due to lower participation and knowledge, public policy still tends to reflect their preferences, particularly in areas like redistribution and government programs.
- The study reviews existing literature and analyzes data on voter participation, political knowledge, and policy outcomes to assess the impact of income inequality on democratic responsiveness.

- **Findings**
 - Despite the lower levels of political participation among low-income voters, Erikson's study finds that public policies often reflect the preferences of this group. This is particularly evident in areas related to social welfare, redistribution, and government assistance programs, where low-income Americans tend to favor stronger governmental intervention.
 - The study also underscores the influence of high-income groups on policy-making. These groups have both the financial resources and organizational infrastructure to significantly influence the legislative process. As a result, policies such as tax cuts and economic deregulation are often tailored to benefit wealthier segments of society.
 - While the wealthiest certainly have the most political pull, the study suggests that public opinion still holds significant sway in shaping public policy.

- **Connection to 1/3: All Thirds**
 - Low-Income America: Even though they may not have the same level of influence as wealthier groups, their preferences still shape policy decisions when there is mass public support. This is particularly true in the domains of redistributive policy and government aid.
 - Middle-Income America: For this group, the findings suggest that their policy demands often require balancing the needs of both higher-income groups and lower-income groups. They also experience a slower pace of policy changes.
 - High-Income America: For this group, the findings suggest that policy changes benefiting them are often more incremental and slow to enact. Policies that promote economic growth in wealthier segments of society are more likely to be implemented, but with potential resistance from lower-income groups and public demand for welfare.

Indicator: Political Power

Intersectional Inequalities: How Socioeconomic Well-Being Varies at the Intersection of Disability, Gender, Race-Ethnicity, and Age

Laurin E. Bixby

Research in Social Stratification and Mobility

June, 2024

- This study examines how poverty and unemployment are distributed across groups defined by the intersection of disability, gender, race-ethnicity, and age. It highlights the complex, compounding effects of multiple systems of oppression—like ableism, racism, sexism, and ageism—on socioeconomic outcomes in the U.S.
- The author argues that disability is a core and under-acknowledged axis of stratification that powerfully interacts with other marginalized identities. Socioeconomic disadvantage is not simply a function of individual limitations but the result of entrenched structural inequalities. Policies and practices that treat these forms of oppression in isolation miss the true scope of inequality and cannot meaningfully improve outcomes.
- Using data from the American Community Survey (2008–2019), covering over 19 million working-age adults, the study used intersectional logistic regression models to estimate poverty (via the Supplemental Poverty Measure) and unemployment across 40 groups defined by disability, gender, race-ethnicity, and age (25–44 vs. 45–64). The study also conducted supplementary analysis across types of disability.
- **Findings**
 - Disability is the strongest axis of inequality for both poverty and unemployment.
 - Disabled Black and AIAN (American Indian/Alaska Native) men and women, especially in middle age, face the highest levels of poverty and unemployment.
 - Within racial and gender categories, disabled people consistently fare worse than their nondisabled counterparts.
 - The groups with the lowest socioeconomic disadvantage are nondisabled White and PI men in early adulthood.
 - Disability-based disparities are wider among men than women and are compounded by race and age, especially among racially minoritized groups.
- **Connection to 1/3s: Bottom Third**
 - For the Bottom Third, especially those who are multiply marginalized (e.g., disabled Black and AIAN individuals), the findings reveal a critical need to center intersectionality in policymaking. Income, race, disability, and age cannot be siloed—programs targeting poverty and joblessness must address structural ableism, improve access to competitive employment, and reform public benefit programs like SSI and Medicaid. Policies must empower—not punish—disabled people, particularly those bearing the compounded burdens of systemic inequality.

Indicator: Social Equity

Transiting to Work: The Role of Private Transportation for Low-Income Households

Garasky, S., Cynthia, N. F., & Jensen, H. H.

The Journal of Consumer Affairs

2006

- This study examines how public transportation availability and quality influence employment outcomes for low-income urban residents. It seeks to understand whether public transit access supports upward mobility by improving labor market participation and job access, especially in disadvantaged communities.
- The authors argue that a lack of reliable, efficient public transportation is a significant barrier to employment for low-income individuals. Transit infrastructure plays a critical role in shaping geographic opportunity and labor market inclusion, particularly for those without access to private vehicles or who live in transit-poor neighborhoods.
- The study uses a combination of spatial analysis, case studies, and empirical data linking transit availability to employment metrics. It examines how transit accessibility (measured by proximity, frequency, and connectivity) relates to job access, commute times, and employment rates across several U.S. metropolitan areas, focusing on economically vulnerable populations.
- **Findings**
 - Improved transit access is positively associated with higher employment rates among low-income individuals, especially in areas with robust and reliable service.
 - Job accessibility via transit significantly affects labor force participation, particularly for those in entry-level or hourly jobs.
 - Transit deserts—areas with poor or no public transportation—are correlated with lower employment outcomes and longer unemployment spells. Policy interventions (e.g., expanded routes, reduced fares, improved service reliability) have shown measurable benefits in improving job access and economic stability for transit-dependent populations.
- **Connection to 1/3s: Bottom Third**

For the Bottom Third, **transit access is a foundational enabler of economic opportunity**. In many Bottom Third communities, inadequate or unreliable public transportation systems effectively limit access to employment, training, and economic mobility. This study underscores that addressing transit inequality is not merely a transportation issue—it is a labor market and equity issue. Investment in affordable, high-frequency, and well-connected transit can significantly improve economic outcomes for Bottom Third households.

Indicator: Transportation Access

Appendix A: Indicators

Table A1: Indicators for Economic Security and Opportunity

Indicator	Definition
Access to housing	The ability of individuals and families to obtain safe, stable housing without paying more than 30 percent of gross income for housing costs, including utilities.
Economic mobility	The capacity of individuals, families, or groups to alter their economic status over a lifetime and across generations. Economic mobility reflects the potential for changes in income, wealth, or economic standing, as well as the ability to create or access new economic pathways, highlighting the openness of systems that shape prosperity and equity across population segments.
Employment	The share of the working-age population (typically ages 16–64) that is currently employed. This measure reflects both labor market participation and job availability, serving as a core indicator of workforce capacity and engagement.
Income levels	The total regular income an individual receives before deduction, —such as taxes and social insurance contribution, —a representing their gross earning capacity.
Infrastructure quality	The condition, functionality, and resilience of physical systems that underpin modern society, such as transportation networks, utilities, broadband infrastructure, and public facilities. Infrastructure quality reflects the structural integrity and performance of the built environment that enables economic and civic life.
Poverty	Poverty represents insufficient income or resources to meet basic needs such as housing, food, healthcare, and education. Poverty reflects material deprivation and economic insecurity, often shaped by structural factors and persistent across generations.
Social mobility	The ability of individuals or groups to change their social status or position within a society over time. Social mobility reflects the presence of opportunities for movement within society's structure, encompassing recognition, prestige, or affiliation, often linked to education, occupation, and cultural capital.

Table A2: Indicators for Health and Human Development

Indicator	Definition
Child well-being	Child well-being is the holistic state of a child's physical health, emotional security, cognitive development, and social inclusion, shaped by the conditions that promote growth and development.
Cultural enrichment	Cultural enrichment is the process by which individuals and communities engage with, express, and preserve diverse cultural traditions, arts, values, and identities, contributing to a shared sense of belonging.
Educational attainment	The cumulative knowledge, skills, and competencies acquired by individuals across a lifespan. Educational attainment reflects the degree to which learning translates into meaningful application, shaping personal development, social contribution, and collective capacity.
Healthcare access and quality	The extent to which individuals can obtain timely, appropriate, and effective medical care. This includes the availability, affordability, and responsiveness of services, as well as the quality of care received and its impact on health outcomes.
Life expectancy	The average length of time individuals are expected to live, along with the extent to which those years are lived in good health and with personal autonomy.
Lifelong learning	The ongoing process through which individuals acquire and apply new knowledge, skills, and competencies across all stages of life. Lifelong learning reflects both personal engagement and the quality, accessibility, and responsiveness of systems that support continuous learning in formal and informal settings.
Mental and physical wellness	Mental and physical wellness refers to the interconnected states of emotional, psychological, and physical health that enable individuals to function effectively.
Quality of education	Quality of education is the extent to which education systems effectively equip learners with relevant knowledge, critical thinking skills, and socio-emotional development.

Table A3: Indicators for Equity, Sustainability, and Community Resilience

Indicator	Definition
Access to technology	The ability of individuals and communities to obtain and effectively use digital devices, the internet, and cellular connectivity, and the necessary digital skills and support to participate fully in educational, economic, civic, and social life
Access to water and food	The ease with which an individual or group can obtain clean water and healthy food.
Civic engagement	The extent to which individuals and groups participate in the processes that shape their communities. Civic engagement reflects the depth, inclusiveness, and consistency of public involvement in local decision-making, from formal institutions to grassroots initiatives.
Climate change resilience	Climate resilience refers to a community's ability to anticipate, respond to, and recover from climate-related hazards while minimizing long-term harm to societal well-being, economic stability, and environmental health.
Community inclusiveness	The recognition, representation, and active participation of people from varied backgrounds, identities, and experiences in all aspects of community life.
Community safety	The conditions and systems that protect individuals from harm, promote justice, and foster trust, well-being, and a shared sense of security in both public and private spaces. Community safety strengthens social bonds and enhances a collective sense of belonging.
Living environment	The physical and ecological conditions that define where people live and how those places influence their daily experience. Living environment reflects how the design, stability, and surrounding context of a place shape exposure to harm, access to opportunity, and the conditions for a healthy and secure life.
Political power	The ability of individuals or groups to access, influence, or participate in political decision-making. This includes formal representation, civic engagement, and the capacity to shape policy or institutional outcomes.
Social equity	Social equity is the fair and just treatment of all people, regardless of their race, ethnicity, sexual orientation, gender identity, creed, status, and background, ensuring equal opportunities and outcomes for all.
Transportation access	The ease with which individuals can move within and between communities, cities, regions, and across national borders. Transportation access reflects the availability, efficiency, and inclusiveness of mobility systems that enable participation in economic, social, and civic life.